

ITWeb Security Summit

Hybrid Cyber / FIMI Incidents

Global Classification by Continent

Incidents with % fit ≥ 0.7 · Six continents

Taxonomy: Cyber-enabled influence · Influence-enabled cyber · Parallel operations

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This document includes all incidents with a % fit score of 0.7 or above. Entries marked *n/a* represent continents where no qualifying incidents were documented in the source dataset.

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Europe

This table classifies 40 incidents targeting European entities according to the cyber–influence taxonomy:

- **(1) Cyber-enabled influence** — cyber operations serving an influence goal (1a–1h)
- **(2) Influence-enabled cyber** — disinformation/deception enabling a cyber intrusion (2a–2h)
- **(3) Parallel cyber–influence** — simultaneous, independent cyber and influence operations by the same actor

Doppelganger / Overload / RT-mirror / Storm-1516 / R-FBI operations all rely on cyber-enabled fake or spoofed media infrastructure (impersonated outlets, look-alike domains, redirect chains, mirror sites). They are classified under **1c (hack-and-post / propaganda injection via cyber-controlled fake assets and impersonation infrastructure)** with **2d (website impersonation)** as the standard secondary type.

- **1.0** — textbook fit (defining act of the sub-type clearly present).
- **0.7–0.9** — strong fit, partial or mixed on one element.
- **≤ 0.5** — capped here for Type-1 incidents without a real unauthorised intrusion or DDoS, and for Type-2 incidents lacking a tactically-connected subsequent intrusion or DDoS by the same actor.
- **Type-3** scored on whether the cyber and influence components are by the same threat actor and close in time. EU-3 (PRC wind-turbine espionage) drops to 0.4 because the "parallel" influence track is loose geopolitical framing rather than a contemporaneous campaign by the same actor.

Incidents included (% fit ≥ 0.7): 13 of 40

ID	Incident	Country/Scope	Primary Type	Secondary Type(s)	Perpetrator	Victim Category	% Fit
EU-1	2024 European Parliament election hybrid interference environment	European Union (27 member states)	3 – parallel cyber-influence (sustained FIMI plus low-impact DDoS and infrastructure pre-positioning)	1c – CIB and media impersonation; 1g – DDoS against transport / govt / health sites	Russia/Russian-linked (EEAS and Commission reporting)	Elections, voters, government, critical infrastructure	0.8
EU-2	Bundestag hack and later influence exploitation	Germany	1b – hack-and-leak (exfiltrated parliamentary data later exploited for influence pressure)	2b – spearphishing / credential theft	Russia/Russian-linked (APT28 / GRU)	Government, parliament, democratic discourse	0.7
EU-15	Gamaredon phishing campaign against Ukrainian government agencies with war-themed lures	Ukraine (with EU policy spillover)	2b – phishing via email (war-themed lures with malware-laced attachments)	3 – parallel cyber-influence (sat within wider Russian hybrid pressure on Ukraine wartime communications)	Russia/Russian-linked (Gamaredon)	Government, wartime institutions	1.0
EU-18	Ghostwriter activity targeting the Baltic region	Lithuania, Latvia, Estonia	1c – hack-and-post (false stories published through compromised or spoofed news websites and politicians' accounts)	2b – phishing / credential theft; 2d – spoofed websites	Russia/Russian-linked and Belarus-linked (Ghostwriter)	Government, media, public discourse, NATO trust	0.9
EU-19	Ghostwriter intrusions targeting German politicians	Germany	1b – hack-and-leak / hack-and-post against Bundestag and Landtag members	2b – phishing / credential compromise	Russia/Russian-linked (Ghostwriter)	Elections, politicians, government	0.8
EU-21	InvisiMole / Gamaredon activity against Ukrainian state bodies	Ukraine	2b – phishing-led intrusion (malicious lures deploying backdoor / surveillance malware)	3 – parallel cyber-influence (espionage paired with wider Russian influence pressure on Ukrainian narratives)	Russia/Russian-linked (InvisiMole tied to Gamaredon)	Government, wartime institutions	1.0
EU-22	Macron Leaks Operation	France	1b – hack-and-leak (timed release of stolen campaign material mixed with forged documents)	2b – phishing / email compromise; 2g – forged material mixed with authentic data to mislead	Russia/Russian-linked (APT28-linked activity)	Elections, presidential campaign, voters	1.0

ID	Incident	Country/Scope	Primary Type	Secondary Type(s)	Perpetrator	Victim Category	% Fit
EU-23	Moldova presidential election and referendum interference campaign	Moldova	3 – parallel cyber-influence (cyberattacks against election infrastructure combined with coordinated disinformation)	1c – CIB; 1g – DDoS against election-related systems	Russia/Russian-linked interference actors	Elections, referendum, government	0.8
EU-33	Russian-linked cyberattacks and information interference during the 2024 European election period in Bulgaria	Bulgaria	3 – parallel cyber-influence (cyberattacks by Russian hacker groups alongside media impersonation and inauthentic amplification)	1c – CIB; 2d – inauthentic news outlets	Russia/Russian-linked actors (per European Commission reporting)	Elections, government, voters	0.7
EU-35	Russian targeting of EU government agencies using war-themed phishing lures	EU member states	2b – phishing via email with war-themed lures and espionage malware	3 – parallel cyber-influence (lure themes aligned with wider Russian influence efforts on the conflict)	Russia/Russian-linked (Gamaredon-related activity)	Government, EU institutions	1.0
EU-37	TV5Monde cyberattack and propaganda misuse	France	1c – hack-and-post / hostile takeover of broadcaster channels (website, social-media accounts, broadcast systems) to post propaganda	1d – hack-and-destroy element (heavy disruption of broadcast services); 2g – false flag ("CyberCaliphate" cover for Russia-linked activity)	Russia/Russian-linked (initial false-flag claim by "CyberCaliphate")	Media, broadcaster, public opinion	1.0
EU-39	Ukrainian IT Army targeting of Russian and Belarusian websites as a cyber-enabled information campaign	Russia, Belarus	1g – DDoS-for-suppression (denying access to government, media, finance, energy, transport and political-party sites)	1f – website effects via service disruption; 3 – parallel cyber-influence (publicised cyber actions used as wartime narrative signal); 2g – use of compromised Docker honeypots as cover	Ukraine / IT Army and affiliated volunteer cyber actors	Government, media, financial sector, critical infrastructure (Russian and Belarusian)	0.9
EU-40	NoName057(16) DDoS campaign against European banks and financial-sector targets	Italy, Czechia, Lithuania, Denmark, Germany, Sweden, Netherlands, Spain, France, Switzerland, plus North American banks	1g – DDoS-for-suppression (denying online-banking and financial-sector availability as political signalling)	1c – cyber-enabled influence via Telegram-claimed attacks framed as Ukraine-war retaliation; 3 – parallel cyber-influence at moments of political symbolism (Czech presidential election, German tank decisions, Ukraine Peace Summit)	NoName057(16) Russia-affiliated "patriotic hacktivist" group with DDoSia tool and ~4,000 volunteers	Private sector (banking and financial services)	0.9

Continental observations

- **Europe is the most concentrated theatre in the dataset (40 of 101 incidents) and the testbed for almost every taxonomy sub-type**, particularly Russia-linked operations connected to Ukraine, EU enlargement, and elections.
- **Five overlapping Russian influence "franchises" dominate**: Doppelganger (EU-6, 7, 8, 9, 10, 17, 20, 32), Operation Overload (EU-4, 24, 25, 26), RT Deutsch mirrors / laundering (EU-30, 31), Ghostwriter (EU-18, 19), and Storm-1516 / R-FBI (EU-29, 36). All share the same primary signature: 1c propaganda injection via cyber-controlled fake assets, with 2d website impersonation as the standard supporting layer.
- **Hack-and-leak (1b) remains rare but high-impact**: Bundestag (EU-2), Ghostwriter against German politicians (EU-19), Macron Leaks (EU-22). Each was preceded by phishing (2b).
- **Type-3 parallel attacks cluster around elections and wartime pressure**: European Parliament 2024 (EU-1), Romania (EU-11), Moldova (EU-23), Bulgaria (EU-33), Poland (EU-28), farmers' protests in Germany (EU-14), and the broader Gamaredon-driven Ukrainian environment (EU-15, 21, 35).
- **The Ukrainian IT Army (EU-39) is the dataset's clearest example of patriotic hacktivism** operating as a cyber-enabled information campaign — particularly relevant for the keynote's question on this convergence.
- **NoName057(16) (EU-40) is the corporate-target mirror image of the same phenomenon** — Russia-aligned patriotic hacktivism using DDoS against European banks as a multi-year political-signalling campaign, eventually disrupted by Europol's Operation Eastwood in July 2025.

- **TV5Monde (EU-37)** is the textbook hack-and-post / hack-and-destroy hybrid combined with a 2g false-flag cover (CyberCaliphate).
- **Western-government covert operations also appear in Europe** (EU-27, Pentagon Kontur Novosti targeting Russian-speaking Ukrainians) — useful nuance for an "all sides" framing in the keynote.

EU-1

2024 European Parliament election hybrid interference environment

% fit: 0.8

Date	2024
Targets	European Parliament elections, EU voters, election information space
Continent	Europe
Perpetrator	Primarily Russian FIMI actors and associated cyber-enabled operations according to EEAS and Commission reporting
Motivation	Undermine trust in the EU, weaken support for Ukraine, and shape electoral behavior

Cyberattack Details

- **Nature:** Cyber-enabled interference, including infrastructure setup ahead of the vote and low-impact DDoS activity in some member states
- **Systems Attacked:** Websites of public organizations and election-related online environments; observed targets included transport, government, and healthcare-related sites
- **Data Held:** Public-service and institutional web systems; exact datasets not enough info
- **Availability Impact:** Very limited number of low-impact DDoS incidents reported
- **Data Exfiltration/Alteration:** Not enough info on exfiltration volumes
- **Estimated Impact:** Persistent election-security pressure rather than major operational disruption
- **Investigators:** EEAS, European Commission services, ENISA, national authorities
- **Remediation:** Monitoring, platform action, stress-testing, election preparedness and public awareness

Disinformation Details

- **Nature:** Sustained Russian FIMI activity before, during, and after the European elections
- **Narratives:** Anti-EU, anti-Ukraine, anti-establishment, abstention-promoting, and trust-undermining narratives
- **Deception:** Bot networks, coordinated inauthentic behavior, impersonation of media outlets, fake websites
- **Platforms:** X, Facebook, websites, other platforms
- **Reach:** Union-wide
- **Impact:** Ongoing pressure on election integrity and trust in results
- **Disinfo Investigators:** EEAS, European Commission, EUvsDisinfo, partner researchers
- **Remediation:** Platform enforcement, exposure, fact-checking, institutional preparedness
- **Link to Cyberattack:** The operation combined cyber-enabled infrastructure, limited disruptive activity, and large-scale influence operations

EU-2

Bundestag hack and later influence exploitation

% fit: 0.7

Date	2015 and after
Targets	German Bundestag, parliamentary communications, German democratic discourse
Continent	Europe
Perpetrator	APT28 / GRU attributed by German authorities
Motivation	Espionage, strategic access to political communications, and influence leverage

Cyberattack Details

- **Nature:** Spearphishing, credential theft, persistent intrusion, data exfiltration
- **Systems Attacked:** Bundestag parliamentary network and email systems
- **Data Held:** Parliamentary emails and internal communications
- **Availability Impact:** Network systems had to be taken offline during remediation
- **Data Exfiltration/Alteration:** Significant data theft publicly reported; exact volume varies by source
- **Estimated Impact:** Major institutional compromise with long-term democratic-security effects
- **Investigators:** German domestic intelligence, federal investigators, partner agencies
- **Remediation:** Network rebuild, public attribution, legal actions, stronger cyber defenses

Disinformation Details

- **Nature:** Broader influence exploitation and propaganda pressure around stolen political data and anti-government narratives
- **Narratives:** Anti-government, anti-migrant, anti-Western, and destabilizing narratives around German politics
- **Deception:** Misleading narratives, fear around stolen data, hybrid pressure on later election cycles
- **Platforms:** Social media, propaganda outlets, online channels
- **Reach:** National political discourse
- **Impact:** Erosion of trust in democratic institutions and concerns over electoral manipulation
- **Disinfo Investigators:** BfV, researchers, journalists, policy analysts
- **Remediation:** Public warnings, election-security measures, counter-disinformation monitoring
- **Link to Cyberattack:** The cyber intrusion created intelligence and coercive material that fit a wider influence context

EU-15

Gamaredon phishing campaign against Ukrainian government agencies with war-themed lures

% fit: 1.0

Date	2022
Targets	Ukrainian government agencies and, by extension, the European policy environment dealing with the war
Continent	Europe
Perpetrator	Russian threat actor Gamaredon
Motivation	Espionage, wartime intelligence gathering, and pressure on institutions central to public and strategic messaging

Cyberattack Details

- **Nature:** Phishing campaign with malware-laced attachments and espionage tooling
- **Systems Attacked:** Government email systems and agency endpoints
- **Data Held:** Government communications and institutional data
- **Availability Impact:** No major public outage reported
- **Data Exfiltration/Alteration:** Malware enabled theft of information from targeted agencies
- **Estimated Impact:** Exposure of government information and increased wartime institutional insecurity
- **Investigators:** Cisco Talos, CERT-UA, and cyber-incident compilers
- **Remediation:** Detection, endpoint defense, phishing awareness, and incident response

Disinformation Details

- **Nature:** The phishing campaign used conflict-related lures tied to the Russia-Ukraine war and sat within a broader Russian hybrid strategy mixing cyber and information pressure
- **Narratives:** War-crime and military-assistance themed messaging used as lure content; broader surrounding ecosystem included anti-Ukrainian and manipulative war narratives
- **Deception:** Social engineering through themed documents and misleading contextual framing
- **Platforms:** Email and malware-delivery infrastructure
- **Reach:** Governmental and strategic audiences
- **Impact:** Increased vulnerability of institutions central to wartime communications
- **Disinfo Investigators:** Security researchers and policy analysts
- **Remediation:** Threat intelligence sharing and awareness efforts
- **Link to Cyberattack:** The cyber compromise was initiated through manipulative information lures connected to the broader conflict narrative space

EU-18 Ghostwriter activity targeting the Baltic region

% fit: 0.9

Date	2020-2022
Targets	Baltic politicians, institutions, media audiences, regional public discourse
Continent	Europe
Perpetrator	Belarus- and Russia-linked Ghostwriter activity in public reporting
Motivation	Influence regional security narratives, weaken trust in NATO and democratic institutions

Cyberattack Details

- **Nature:** Account compromise, phishing, website compromise, credential theft
- **Systems Attacked:** Politicians' accounts, websites, communications systems
- **Data Held:** Account access, correspondence, web content
- **Availability Impact:** Limited direct outages publicly reported
- **Data Exfiltration/Alteration:** Website and account content manipulated; exfiltration scope not enough info
- **Estimated Impact:** Regional information manipulation with political and security implications
- **Investigators:** Security researchers, national authorities, journalists
- **Remediation:** Site restoration, public attribution, stronger account security

Disinformation Details

- **Nature:** Fabricated stories and false posts published through compromised or spoofed channels
- **Narratives:** NATO misconduct, anti-Western themes, security alarmism
- **Deception:** Hacked or spoofed websites and personas used to launder false narratives
- **Platforms:** News websites, blogs, social media
- **Reach:** Baltic and wider European audiences
- **Impact:** Trust erosion and regional insecurity
- **Disinfo Investigators:** Researchers and national authorities
- **Remediation:** Takedowns, public exposure, attribution
- **Link to Cyberattack:** The false narratives were distributed via compromised or spoofed digital infrastructure

EU-19

Ghostwriter intrusions targeting German politicians

% fit: 0.8

Date	2021
Targets	Bundestag members, state parliament members, German election information environment
Continent	Europe
Perpetrator	Russia-linked Ghostwriter activity according to German government warnings and public reporting
Motivation	Influence the federal election, gather political intelligence, and shape public discourse

Cyberattack Details

- **Nature:** Phishing and credential compromise against politicians
- **Systems Attacked:** Email accounts and digital communications of elected officials
- **Data Held:** Political correspondence and account access
- **Availability Impact:** No major service outage publicly reported
- **Data Exfiltration/Alteration:** Intrusions and attempted access reported; full exfiltration scope not enough info
- **Estimated Impact:** Election-security risk and possible exposure of political communications
- **Investigators:** German authorities and partner agencies
- **Remediation:** Public warning, stronger account security, election cyber monitoring

Disinformation Details

- **Nature:** Election-period influence activity associated with fake personas and content laundering
- **Narratives:** Content meant to embarrass politicians and weaken trust in institutions
- **Deception:** Impersonation, false personas, and manipulated messaging
- **Platforms:** Email, websites, social media
- **Reach:** German media and political ecosystems
- **Impact:** Heightened concern over election interference
- **Disinfo Investigators:** German officials, journalists, researchers
- **Remediation:** Exposure of the campaign and security guidance for political actors
- **Link to Cyberattack:** The information manipulation relied on cyber intrusions and account compromise

EU-21

InvisiMole / Gamaredon activity against Ukrainian state bodies

% fit: 1.0

Date	2022
Targets	Ukrainian state bodies and the wartime information environment around them
Continent	Europe
Perpetrator	Russian-linked InvisiMole activity tied in public reporting to Gamaredon
Motivation	Espionage, surveillance, and hybrid pressure against Ukrainian state institutions

Cyberattack Details

- **Nature:** Phishing, backdoor deployment, surveillance malware installation
- **Systems Attacked:** Ukrainian state-body computers and agency infrastructure
- **Data Held:** Government data and internal communications
- **Availability Impact:** No major public outage reported
- **Data Exfiltration/Alteration:** Surveillance tooling allowed collection and follow-on compromise; exact exfiltration volumes not enough info
- **Estimated Impact:** Deepened exposure of Ukrainian government systems during wartime
- **Investigators:** CERT-UA, ZDNet reporting, and cyber-incident compilers
- **Remediation:** Detection, cleanup, incident response, threat intelligence sharing

Disinformation Details

- **Nature:** The operation fit a hybrid environment in which cyber intrusions supported a broader Russian effort to manipulate narratives and weaken Ukrainian institutional credibility
- **Narratives:** Anti-Ukrainian and war-related manipulation in the surrounding information space
- **Deception:** Malicious lures, covert access, and strategic timing in a conflict setting
- **Platforms:** Email and endpoint infrastructure; broader surrounding narrative distribution via media and online channels
- **Reach:** Governmental and strategic
- **Impact:** Heightened institutional vulnerability amid intensive information warfare
- **Disinfo Investigators:** Security researchers and conflict analysts
- **Remediation:** Public warnings and resilience measures
- **Link to Cyberattack:** The espionage activity supported broader hybrid pressure on Ukraine's wartime state communications and narrative environment

EU-22

Macron Leaks Operation

% fit: 1.0

Date	April-May 2017
Targets	Emmanuel Macron presidential campaign, French voters, French election information space
Continent	Europe
Perpetrator	Suspected Russia-linked actors; hack attributed by multiple researchers to APT28-linked activity and amplified by pro-Kremlin and alt-right networks
Motivation	Influence the French presidential election and damage Macron's credibility

Cyberattack Details

- **Nature:** Phishing, email compromise, data theft, timed leak of stolen material
- **Systems Attacked:** Campaign email accounts and campaign IT systems
- **Data Held:** Internal emails, campaign files, contracts, financial and operational documents
- **Availability Impact:** No major long-term outage publicly reported
- **Data Exfiltration/Alteration:** Stolen files were leaked; public reporting also noted forged material mixed with authentic content
- **Estimated Impact:** Electoral interference attempt and reputational harm during the final campaign period
- **Investigators:** French authorities, journalists, and DFRLab researchers
- **Remediation:** Public rebuttal, rapid incident response, media restraint during election blackout, post-incident analysis

Disinformation Details

- **Nature:** Coordinated amplification of hacked material with forged and misleading claims
- **Narratives:** Macron as corrupt, compromised, or unfit for office
- **Deception:** Mixture of authentic stolen files and forged documents
- **Platforms:** Twitter/X, 4chan, Pastebin, leak-amplification channels, other social platforms
- **Reach:** International online reach focused on the French election
- **Impact:** Attempted to pollute the election information space and reduce trust in the candidate
- **Disinfo Investigators:** Atlantic Council DFRLab, journalists, academic researchers
- **Remediation:** Fact-checking, campaign rebuttal, limited mainstream amplification
- **Link to Cyberattack:** The disinformation campaign was built directly on the hack-and-leak operation

EU-23

Moldova presidential election and referendum interference campaign

% fit: 0.8

Date	2024
Targets	Moldovan election infrastructure, referendum process, political discourse, Moldovan voters
Continent	Europe
Perpetrator	Russia-linked interference actors according to Moldovan and EU public reporting
Motivation	Influence election outcomes, weaken pro-EU alignment, destabilize democratic processes

Cyberattack Details

- **Nature:** Cyberattacks against election-related systems combined with broader digital interference
- **Systems Attacked:** Election-related IT systems and associated digital infrastructure
- **Data Held:** Election-related administrative and political data; exact data categories not enough info
- **Availability Impact:** Not enough info on major outages, but incidents triggered international support measures
- **Data Exfiltration/Alteration:** Not enough info on confirmed exfiltration volumes
- **Estimated Impact:** Attempted disruption of election integrity and public trust
- **Investigators:** Moldovan authorities, OSCE observers, EU institutions, Cyber Rapid Response support
- **Remediation:** Election readiness exercises, platform coordination, public warnings, international support

Disinformation Details

- **Nature:** Coordinated disinformation, influence activity, and social media manipulation around the election
- **Narratives:** Anti-EU claims, election fraud narratives, and false accusations linked to the vote
- **Deception:** Coordinated false content and manipulative amplification
- **Platforms:** Major social platforms and influence networks
- **Reach:** National reach with regional and international attention
- **Impact:** Pressure on voter confidence and perceived legitimacy
- **Disinfo Investigators:** OSCE, EU institutions, platforms, civil society, Moldovan authorities
- **Remediation:** Fact-checking, monitoring, takedowns, rapid coordination channels
- **Link to Cyberattack:** Public reporting describes a combined pattern of cyberattacks and disinformation against the electoral process

EU-33

Russian-linked cyberattacks and information interference during the 2024 European election period in Bulgaria

% fit: 0.7

Date	2024
Targets	Bulgarian institutions tied to the election period, Bulgarian public discourse, broader EU election integrity
Continent	Europe
Perpetrator	Russian hacker groups and associated interference networks according to European Commission reporting
Motivation	Undermine trust in elections, create disruption, and support pro-Kremlin narratives

Cyberattack Details

- **Nature:** Cyberattacks by Russian hacker groups during the electoral period
- **Systems Attacked:** Bulgarian institutional systems; precise systems not enough info in the cited memo
- **Data Held:** Not enough info
- **Availability Impact:** Not enough info
- **Data Exfiltration/Alteration:** Not enough info
- **Estimated Impact:** Election-security pressure and disruption risk
- **Investigators:** Bulgarian authorities, European Commission services, election-security networks
- **Remediation:** Election monitoring, platform cooperation, EU-level countermeasures

Disinformation Details

- **Nature:** Information interference tied to the 2024 EU elections, including media impersonation and inauthentic amplification
- **Narratives:** Anti-Ukraine, anti-EU, anti-migrant, and polarizing election messaging
- **Deception:** Media impersonation, inauthentic news outlets, and coordinated false accounts
- **Platforms:** X, Facebook, websites, coordinated networks
- **Reach:** Multi-country EU reach including Bulgaria
- **Impact:** Manipulation of election-related discourse
- **Disinfo Investigators:** EEAS, EDMO, European Commission services, national authorities
- **Remediation:** Takedowns, monitoring, and election stress-testing
- **Link to Cyberattack:** The reported election threat environment combined cyberattacks and information interference

EU-35

Russian targeting of EU government agencies using war-themed phishing lures

% fit: 1.0

Date	2022
Targets	Government agencies of EU member states
Continent	Europe
Perpetrator	Russian Federation-linked threat actors, associated in reporting with Gamaredon-related activity
Motivation	Espionage, strategic access, and pressure on European institutions during the war in Ukraine

Cyberattack Details

- **Nature:** Phishing campaign with malicious attachments and espionage malware
- **Systems Attacked:** Government-agency email systems and endpoints
- **Data Held:** Official communications and agency documents
- **Availability Impact:** No major public outage reported
- **Data Exfiltration/Alteration:** Malware enabled covert access and likely information theft
- **Estimated Impact:** Compromise risk for public institutions across the EU
- **Investigators:** Security researchers and public cyber defenders
- **Remediation:** Email filtering, patching, endpoint monitoring, phishing-awareness measures

Disinformation Details

- **Nature:** The phishing lures exploited war-related informational themes and aligned with wider Russian influence efforts around the conflict
- **Narratives:** War crimes, military assistance, and conflict framing designed to induce engagement with malicious material
- **Deception:** Social-engineering documents that disguised malware as relevant wartime information
- **Platforms:** Email and malicious attachment infrastructure
- **Reach:** Multiple EU member-state agencies
- **Impact:** Increased susceptibility of EU institutions to hybrid wartime manipulation
- **Disinfo Investigators:** Security analysts and policy researchers
- **Remediation:** Alerts and awareness measures
- **Link to Cyberattack:** The cyber intrusion depended on manipulative informational lures embedded in the phishing operation

EU-37 TV5Monde cyberattack and propaganda misuse

% fit: 1.0

Date	April 2015
Targets	TV5Monde broadcast systems, website, social media accounts, French-speaking audiences
Continent	Europe
Perpetrator	Attack initially claimed by the "CyberCaliphate"; later public reporting linked the operation to Russian actor activity
Motivation	Disrupt broadcasting and exploit compromised media channels for propaganda and intimidation

Cyberattack Details

- **Nature:** Network intrusion, account compromise, service disruption, hostile takeover of broadcaster channels
- **Systems Attacked:** Broadcast management systems, website, social media accounts
- **Data Held:** Operational platform access and media account control
- **Availability Impact:** Broadcast services were heavily disrupted
- **Data Exfiltration/Alteration:** Accounts and content were altered; not enough info on large exfiltration
- **Estimated Impact:** Operational outage, intimidation, and reputational damage
- **Investigators:** French authorities and cybersecurity investigators
- **Remediation:** System rebuilds, access-control hardening, restoration of services

Disinformation Details

- **Nature:** False-flag propaganda posted via compromised broadcaster channels
- **Narratives:** Extremist intimidation and misleading attribution
- **Deception:** Impersonation through hijacked trusted media infrastructure
- **Platforms:** TV channels, website, Facebook, Twitter/X
- **Reach:** Wide French-speaking and international audience
- **Impact:** Panic, confusion, and reduced trust in a trusted broadcaster
- **Disinfo Investigators:** French investigators and later cybersecurity researchers
- **Remediation:** Content removal, channel recovery, stronger security controls
- **Link to Cyberattack:** The propaganda component depended entirely on the cyber compromise

EU-39

Ukrainian IT Army targeting of Russian and Belarusian websites as a cyber-enabled information campaign

% fit: 0.9

Date	2022
Targets	Russian and Belarusian government, media, finance, energy, transport, and political-party websites; European regional information space affected indirectly by the campaign
Continent	Europe
Perpetrator	Ukraine's IT Army and affiliated volunteer cyber actors as reflected in the uploaded cyber-operations dataset
Motivation	Impose pressure on Russian and Belarusian digital targets while amplifying wartime messaging and participation narratives

Cyberattack Details

- **Nature:** DDoS activity and compromise of Docker honeypots to launch pro-Ukrainian attacks
- **Systems Attacked:** Government, media, finance, energy, transport, and political-party websites in Russia and Belarus
- **Data Held:** Public-facing web infrastructure; not enough info on sensitive datasets
- **Availability Impact:** Website outages and service disruption
- **Data Exfiltration/Alteration:** Not enough info on exfiltration; the main effect was availability disruption
- **Estimated Impact:** Disrupted target services and demonstrated mobilized cyber participation
- **Investigators:** CrowdStrike and the dataset compilers
- **Remediation:** Defensive filtering, infrastructure restoration, anti-DDoS measures

Disinformation Details

- **Nature:** The operation sat within a broader wartime information effort in which cyber actions were publicized, coordinated, and used as influence signals
- **Narratives:** Pro-Ukrainian resistance, Russian vulnerability, and participatory digital mobilization
- **Deception:** Use of online coordination channels and information framing around the cyber campaign
- **Platforms:** Telegram and other online coordination spaces; public reporting and social media amplification
- **Reach:** Regional and international
- **Impact:** Reinforced wartime narratives and encouraged digital volunteerism
- **Disinfo Investigators:** Security researchers and wartime information analysts
- **Remediation:** Not enough info on any single counter-disinformation response
- **Link to Cyberattack:** The campaign paired cyber disruption with overt information effects and narrative amplification

EU-40

NoName057(16) DDoS campaign against European banks and financial-sector targets

% fit: 0.9

Date	2022 through 2025 (campaign ongoing; major disruptions across this window; Europol "Operation Eastwood" takedown 16 July 2025)
Targets	European financial-sector corporations — including Italian banks (Banca Intesa Sanpaolo, Banca Monte dei Paschi di Siena, Banca Popolare di Bari, BPER Banca, Iliad Italia), Czech banks (multiple in August 2023), Lithuanian SEB (July 2022), Danish banks (June 2022 and 2023), Canadian Laurentian Bank and TD Bank (September 2023), as well as Swedish bank websites, plus over 250 German entities targeted across 14 attack waves from November 2023 onward
Continent	Europe
Perpetrator	NoName057(16), a Russia-affiliated "patriotic hacktivist" group active since March 2022, operating with approximately 4,000 volunteers via the DDoSia distributed denial-of-service tool and several hundred botnet servers; closely aligned with Killnet, XakNet and other pro-Kremlin operations
Motivation	Political signalling against European supporters of Ukraine — financial-sector disruption serves as a visible pressure mechanism against NATO-aligned states without crossing into the kind of destructive attacks that would invite kinetic retaliation

Cyberattack Details

- **Nature:** Distributed denial-of-service (DDoS) campaigns conducted via the DDoSia tool plus a several-hundred-server botnet; relatively simple technically but persistent, with daily attack waves typically timed to coincide with political events
- **Systems Attacked:** Public-facing websites of European banks and financial institutions, payment gateways, mobile-application back-ends, and other availability-dependent services
- **Data Held:** None — DDoS attacks are availability-focused, not data-theft operations
- **Availability Impact:** Variable — most attacks produce temporary website unavailability for minutes to hours; some attacks against online banking caused service disruption for legitimate customers; Operation Eastwood reporting noted that "most attacks have been mitigated without any substantial interruptions"
- **Data Exfiltration/Alteration:** None
- **Estimated Impact:** Cumulative cost across hundreds of attacks is substantial in mitigation infrastructure and operational disruption; psychological impact on customers using online banking during attack windows; financial-sector regulators across multiple EU states issuing advisories; recognition that "patriotic hacktivism" is now a persistent operational threat to financial services
- **Investigators:** Europol (Operation Eastwood), national CERTs and police forces in Czechia, Finland, France, Germany, Italy, Lithuania, Netherlands, Poland, Spain, Sweden and the US; private-sector threat intelligence from Check Point, Radware, SentinelLabs, Recorded Future and others
- **Remediation:** Operation Eastwood (16 July 2025) — takedown of more than 100 servers and major portions of NoName's central infrastructure; two arrests (France and Spain); six German arrest warrants for Russian nationals including suspected ringleaders; over 1,100 alleged supporters contacted via encrypted-messaging-app warnings; sectoral DDoS protections strengthened

Disinformation Details

- **Nature:** The DDoS attacks themselves function as influence operations — they are messaging acts more than they are technical disruptions, with each attack publicly claimed and contextualised as retaliation for specific Western support for Ukraine
- **Narratives:** "We will continue to attack Western countries that support Ukraine"; specific attacks framed as retaliation for named political decisions (e.g. Italy's G7 reaffirmation of support for Ukraine; Czech Republic training Ukrainian soldiers; Germany sending Leopard 2 tanks); banks framed as legitimate targets because they support sanctioned trade flows
- **Deception:** "Hactivist" rather than state-affiliated branding to maintain plausible deniability for the Russian state; claims of independent volunteer operation despite documented coordination with Killnet and other pro-Kremlin groups
- **Platforms:** Telegram channels (main Russian-language channel with around 20,000 members, English-language equivalent, DDoSia volunteer-recruitment channels), GitHub (DDoSia tooling and configuration files), CheckHost website (for boasting about attack effectiveness)
- **Reach:** Mass — millions of European citizens experienced (often brief) availability issues with their online banking; sustained media coverage across European cybersecurity press
- **Impact:** Demonstration that volunteer-based hacktivism can sustain a multi-year campaign against European critical financial infrastructure; significant policy attention from Europol and national cybersecurity agencies; catalysed cross-border law-enforcement cooperation culminating in Operation Eastwood
- **Disinfo Investigators:** Check Point Research, Radware, SentinelLabs, Recorded Future, Computer Weekly, The Record, CybelAngel, White Blue Ocean; Europol public communications
- **Remediation:** Operation Eastwood takedown; pan-European DDoS-resilience guidance; ongoing monitoring of follow-on hactivist groups
- **Link to Cyberattack:** The influence and the cyberattack are inseparable — each DDoS is itself a political message, with public Telegram claims and CheckHost screenshots forming the influence layer on top of the technical attack

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North America

This table classifies 16 incidents targeting North American entities according to the cyber–influence taxonomy:

- **(1) Cyber-enabled influence** — cyber operations serving an influence goal (1a–1h)
 - **(2) Influence-enabled cyber** — disinformation/deception enabling a cyber intrusion (2a–2h)
 - **(3) Parallel cyber–influence** — simultaneous, independent cyber and influence operations by the same actor
 - **1.0** — textbook fit (defining act of the sub-type clearly present).
 - **0.7–0.9** — strong fit, partial or mixed on one element.
 - **≤ 0.5** — capped here for Type-1 incidents without a real unauthorised intrusion or DDoS, and for Type-2 incidents lacking a tactically-connected subsequent intrusion or DDoS by the same actor.
 - **Type-3** scored on whether the cyber and influence components are by the same threat actor and close in time.
- NA-11 (Russian active measures 2016–2025) drops to 0.5 because the long span of years stretches "parallel" beyond contemporaneous co-occurrence.

Incidents included (% fit ≥ 0.7): 9 of 16

ID	Incident	Country/Scope	Primary Type	Secondary Type(s)	Perpetrator	Victim Category	% Fit
NA-5	Democratic Congressional Campaign Committee intrusion and leak activity	United States	1b – hack-and-leak (thousands of stolen documents surfaced through leak and media channels)	2b – phishing / credential theft; 2g – false-flag personas	Russia/Russian-linked actors (part of the 2016 election interference effort)	Political party, congressional campaigns, elections	1.0
NA-6	Democratic National Committee hack-and-leak operation	United States	1b – hack-and-leak (stolen emails released via Guccifer 2.0, DCLeaks, WikiLeaks)	2b – phishing / credential theft; 2g – false-flag cutout personas (Guccifer 2.0, DCLeaks)	Russia/Russian-linked actors (APT28 / Fancy Bear)	Political party, election ecosystem, voters	1.0
NA-7	IRGC Hack-and-Leak Operation Against the 2024 US Presidential Election	United States	1b – hack-and-leak (stolen non-public Trump-campaign material selectively delivered to media and to Biden-campaign associates)	2b – spearphishing / account compromise; 2g – concealment of Iranian state role	Iran (three IRGC-linked nationals)	Elections, presidential campaigns, media, voters	1.0
NA-9	John Podesta spearphish and leak operation	United States	1b – hack-and-leak (large volumes of email stolen and released via WikiLeaks)	2b – spearphishing of personal account; 2g – stolen data laundered through publishing fronts	Russia/Russian-linked operators	Elections, presidential campaign leadership, voters	1.0
NA-10	(Mexico) No source-supported hybrid incidents met the threshold for inclusion	Mexico	Not applicable (no incidents documented)	—	Not applicable	Not applicable	n/a
NA-13	Sony Pictures "Guardians of Peace" hack-and-leak, wiper deployment and <i>The Interview</i> coercion campaign	United States	1b – hack-and-leak (~100TB exfiltrated, staged data dumps over weeks)	1d – hack-and-destroy (Shamoon-variant wiper); 2g – "Guardians of Peace" false-flag persona masking state attribution	DPRK-linked actors (FBI attribution, December 2014)	Private sector (film studio), employees, executives, downstream cinema chains	1.0
NA-14	MGM Resorts Scattered Spider vishing and ALPHV ransomware attack	United States	2c – deepfake/impersonation-style scam (English-fluent vishing of IT help desk while impersonating a real MGM employee)	2f – doxxing-based (double-extortion) ransomware via ALPHV / BlackCat deployment	Scattered Spider (UNC3944) as ALPHV / BlackCat affiliate	Private sector (hospitality / casino), customers, employees	1.0
NA-15	JBS Foods REvil ransomware attack on global meat-processing operations	United States (primary impact), Canada, Australia	2f – doxxing-based (double-extortion) ransomware	1d – hack-and-destroy (operational shutdown of beef and pork slaughterhouses across three continents)	REvil / Sodinokibi (Russian-speaking RaaS gang, FBI attribution)	Private sector (food production), global meat supply chain	1.0

ID	Incident	Country/Scope	Primary Type	Secondary Type(s)	Perpetrator	Victim Category	% Fit
NA-16	Change Healthcare ALPHV / BlackCat ransomware, exit-scam and RansomHub re-extortion	United States	2f – doxxing-based (double-extortion) ransomware	1d – hack-and-destroy (weeks-long national pharmacy and claims-processing outage); 2g – ALPHV fake "law-enforcement seizure" notice used to disguise exit-scam against its own affiliate	ALPHV / BlackCat affiliate, then RansomHub for re-extortion after ALPHV exit-scam	Private sector (healthcare-payments infrastructure), 100M+ patients, downstream providers	1.0

Continental observations

- **North America is the global capital of hack-and-leak operations (Type 1b).** Six of the original 12 incidents — DNC (NA-6), DCCC (NA-5), Podesta (NA-9), IRGC-2024 (NA-7), Iranian think-tank targeting (NA-8), and cyber-targeting of journalists (NA-4) — fit this category. Every one was preceded or enabled by spearphishing (2b), and three of them deployed cutout personas and laundering channels (2g) to disguise state attribution. Sony Pictures (NA-13) is a corporate-target analogue with both 1b and 1d components plus a textbook 2g false-flag persona.
- **Russia dominates the historic record (DNC, DCCC, Podesta, biolabs, active measures); Iran has emerged as a sustained second perpetrator (IRGC-2024, APT42 think-tank targeting); China is increasingly visible (Canadian elections, US political reporters); and DPRK is responsible for the most dramatic corporate-target case (Sony Pictures).**
- **The corporate cases (NA-13 through NA-16) introduce three distinct profiles:** state-attributed hack-and-leak with destructive intent (Sony), criminal ransomware with influence-style social-engineering entry (MGM), and pure-financial double-extortion ransomware with major societal impact (JBS, Change Healthcare). Change Healthcare additionally features an unusual threat-actor-against-threat-actor deception (ALPHV's fake law-enforcement-seizure exit-scam against its own affiliate).
- **NA-10 (Mexico) is recorded for completeness** — the source file explicitly notes that no incidents met the inclusion threshold; this is not a "gap" but a documented absence in the underlying dataset.
- **NA-1 (Canada 2025) and NA-11 (Russian active measures) are textbook Type-3 hybrid environments** in which authorities explicitly assessed cyber and influence as a single threat picture rather than two separate ones — a useful framing for the keynote.

NA-5

Democratic Congressional Campaign Committee intrusion and leak activity

% fit: 1.0

Date	2016
Targets	Democratic Congressional Campaign Committee, congressional campaign ecosystem, U.S. election discourse
Continent	North America
Perpetrator	Russian intelligence-linked actors connected to the broader 2016 election interference effort
Motivation	Gather politically useful data and amplify narratives harmful to Democratic candidates

Cyberattack Details

- **Nature:** Network intrusion, malware, credential theft, and exfiltration
- **Systems Attacked:** DCCC network and campaign systems
- **Data Held:** Internal campaign documents and congressional race materials
- **Availability Impact:** No major public outage reported
- **Data Exfiltration/Alteration:** Thousands of documents were reportedly stolen and later surfaced through leak and media channels
- **Estimated Impact:** Broader operational and reputational pressure across congressional races
- **Investigators:** U.S. federal investigators, cybersecurity responders, journalists
- **Remediation:** Incident response, network cleanup, public warnings, security upgrades

Disinformation Details

- **Nature:** Leak amplification and narrative shaping around stolen political materials
- **Narratives:** Democratic incompetence, corruption, and election-related distrust
- **Deception:** Use of false personas and selective release to maximize political effect
- **Platforms:** Media outreach, leak sites, social platforms
- **Reach:** U.S. political and media audiences
- **Impact:** Intensified distrust and pressured competitive congressional races
- **Disinfo Investigators:** Journalists, U.S. officials, policy analysts
- **Remediation:** Attribution and tighter campaign cybersecurity
- **Link to Cyberattack:** The narrative campaign relied on stolen data from the network intrusion

NA-6

Democratic National Committee hack-and-leak operation

% fit: 1.0

Date	2015-2016
Targets	Democratic National Committee, U.S. presidential election, U.S. voters
Continent	North America
Perpetrator	Russian intelligence-linked actors, including activity attributed publicly to APT28/Fancy Bear and related personas such as Guccifer 2.0 and DCLeaks
Motivation	Influence the 2016 U.S. presidential election, damage Democratic candidates, and shape U.S. public opinion

Cyberattack Details

- **Nature:** Spearphishing, credential theft, network intrusion, malware deployment, and data exfiltration
- **Systems Attacked:** DNC computer network, email systems, campaign-related accounts
- **Data Held:** Internal emails, opposition research, staff communications, strategic political documents
- **Availability Impact:** No major public-service outage, but network integrity was significantly compromised
- **Data Exfiltration/Alteration:** Stolen emails and documents were exfiltrated and leaked through cutout personas and publishing fronts
- **Estimated Impact:** Major electoral disruption, reputational damage, and long-term loss of trust in democratic processes
- **Investigators:** FBI, private incident responders, U.S. intelligence agencies, Senate investigators
- **Remediation:** Incident response, public attribution, account security improvements, sanctions and legal actions

Disinformation Details

- **Nature:** Coordinated hack-and-leak amplification and narrative manipulation using personas and publication fronts
- **Narratives:** Democratic corruption, election rigging, anti-Clinton narratives, institutional distrust
- **Deception:** Use of false personas such as Guccifer 2.0 and DCLeaks to disguise the state source of stolen information
- **Platforms:** DCLeaks, WikiLeaks, social media, media amplification channels
- **Reach:** National and international
- **Impact:** Distorted election discourse and reduced trust in democratic institutions
- **Disinfo Investigators:** U.S. intelligence agencies, Senate investigators, journalists, researchers
- **Remediation:** Attribution, fact-checking, media scrutiny of leaked content, cyber defense improvements
- **Link to Cyberattack:** The disinformation campaign relied directly on the cyber theft and staged release of stolen files

NA-7

IRGC Hack-and-Leak Operation Against the 2024 US Presidential Election

% fit: 1.0

Date	June-September 2024
Targets	Individuals associated with the Trump campaign, individuals then associated with the Biden campaign, U.S. media organizations, and confidence in the U.S. electoral process
Continent	North America
Perpetrator	Three Iranian nationals linked to the Islamic Revolutionary Guard Corps
Motivation	Stoke discord, erode confidence in the U.S. electoral process, and weaponize stolen campaign material for Iranian strategic objectives

Cyberattack Details

- **Nature:** Spearphishing, account compromise, credential theft, and theft of non-public campaign material followed by unsolicited delivery of stolen excerpts to political and media recipients
- **Systems Attacked:** Personal and campaign-associated online accounts tied to a U.S. presidential campaign, email systems, and media-facing communication channels
- **Data Held:** Non-public campaign material, internal communications, and other sensitive political information associated with the targeted campaign
- **Availability Impact:** No major service outage publicly reported
- **Data Exfiltration/Alteration:** Stolen non-public material was exfiltrated and then selectively leaked by email to political actors and media organizations
- **Estimated Impact:** Attempted election interference, reputational damage, and reduced confidence in campaign and election integrity; U.S. officials described the effort as a direct assault on democratic processes
- **Investigators:** U.S. Department of Justice, FBI, ODNI, and CISA
- **Remediation:** Criminal indictment, public attribution, joint government warnings, and guidance to political organizations on foreign threat activity

Disinformation Details

- **Nature:** Hack-and-leak influence operation using stolen materials as narrative ammunition in the U.S. election environment
- **Narratives:** Discord within the U.S. political system, mistrust of campaigns, and reduced confidence in the fairness and security of the election process
- **Deception:** Spoofed or unsolicited email delivery, covert theft followed by selective leaking, and concealment of the Iranian state role behind the leak activity
- **Platforms:** Email and media dissemination channels
- **Reach:** Targeted political and media recipients; broader potential reach depended on downstream press coverage and public debate
- **Impact:** Sought to shape public opinion and media narratives by laundering hacked material into the election information space
- **Disinfo Investigators:** DOJ, FBI, ODNI, and CISA
- **Remediation:** Public exposure, indictments, election-security advisories, and heightened monitoring of foreign cyber-enabled influence attempts

- **Link to Cyberattack:** The influence component depended entirely on the theft and selective release of hacked campaign material

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NA-9

John Podesta spearphish and leak operation

% fit: 1.0

Date	2016
Targets	Hillary Clinton campaign leadership, U.S. election information space, U.S. voters
Continent	North America
Perpetrator	Russian intelligence-linked operators using spearphishing and cutout dissemination channels
Motivation	Influence the U.S. election and damage the Clinton campaign

Cyberattack Details

- **Nature:** Spearphishing, credential theft, account compromise, data exfiltration
- **Systems Attacked:** Personal and campaign-related email accounts
- **Data Held:** Emails, attachments, campaign communications, internal political strategy discussions
- **Availability Impact:** No known service outage
- **Data Exfiltration/Alteration:** Large volumes of email data were stolen and later released publicly
- **Estimated Impact:** Significant reputational and political impact during the presidential campaign
- **Investigators:** U.S. intelligence agencies, journalists, campaign responders
- **Remediation:** Password resets, public attribution, campaign communications response, wider election security actions

Disinformation Details

- **Nature:** Timed release and narrative amplification of stolen communications
- **Narratives:** Corruption, elite conspiracy, anti-Clinton messaging, distrust of democratic institutions
- **Deception:** Stolen data laundered through publishing fronts and framed for maximum political effect
- **Platforms:** WikiLeaks, social media, partisan media ecosystems
- **Reach:** National and international
- **Impact:** Shaped media cycles and deepened distrust among U.S. voters
- **Disinfo Investigators:** U.S. officials, journalists, academic researchers
- **Remediation:** Attribution, media scrutiny, stronger phishing awareness and cyber protections
- **Link to Cyberattack:** The disinformation effect depended on the successful cyber theft and staged release of emails

NA-10

Not enough source-supported hybrid cyber/FIMI incidents met the threshold for inclusion at this stage

% fit: n/a

Date	not enough info
Targets	Mexican entities in North America
Continent	North America
Perpetrator	not enough info
Motivation	not enough info

Cyberattack Details

- **Nature:** not enough info
- **Systems Attacked:** not enough info
- **Data Held:** not enough info
- **Availability Impact:** not enough info
- **Data Exfiltration/Alteration:** not enough info
- **Estimated Impact:** not enough info
- **Investigators:** not enough info
- **Remediation:** not enough info

Disinformation Details

- **Nature:** not enough info
- **Narratives:** not enough info
- **Deception:** not enough info
- **Platforms:** not enough info
- **Reach:** not enough info
- **Impact:** not enough info
- **Disinfo Investigators:** not enough info
- **Remediation:** not enough info
- **Link to Cyberattack:** not enough info

NA-13

Sony Pictures hack-and-leak, wiper deployment and "Guardians of Peace" coercion campaign

% fit: 1.0

Date	Intrusion ongoing for months prior; public release of stolen data and wiper deployment beginning 24 November 2014; threats relating to <i>The Interview</i> through December 2014
Targets	Sony Pictures Entertainment (SPE), its employees and their families, senior executives Amy Pascal and Michael Lynton, unreleased films, and cinemas planning to screen <i>The Interview</i>
Continent	North America
Perpetrator	A group calling itself "Guardians of Peace" (GOP), publicly attributed by the FBI to North Korea-linked actors in December 2014 (an attribution that has been disputed by some independent cybersecurity experts)
Motivation	Coercion to force Sony Pictures to withdraw <i>The Interview</i> , a satirical comedy depicting the assassination of North Korean leader Kim Jong-un; secondary motives included reputational damage to a major Western media corporation and demonstration of state-aligned cyber capability

Cyberattack Details

- **Nature:** Long-running network intrusion (US investigators say at least two months of file-copying access prior to discovery; alleged GOP members claimed access for a year), followed by massive data exfiltration and deployment of a Shamoon-variant wiper malware
- **Systems Attacked:** Sony Pictures' corporate network and file servers across multiple internal systems; the wiper rendered Sony's computer infrastructure inoperable
- **Data Held:** Personal information about Sony Pictures employees and their families, executive emails (including Amy Pascal's and Michael Lynton's), salary records, copies of then-unreleased Sony films (including *Annie*, *Fury*, *Mr. Turner*), scripts for future films, and other confidential corporate data
- **Availability Impact:** Severe — Sony's computer infrastructure was wiped, forcing extensive operational fallback; the studio retracted *The Interview*'s wide theatrical release in response to threats
- **Data Exfiltration/Alteration:** Up to 100TB allegedly stolen (claim never fully confirmed); multiple staged data dumps released to attract media attention; wiper destruction of source systems
- **Estimated Impact:** Major reputational damage to senior executives (Amy Pascal eventually departed); release of *The Interview* limited to digital and selected independent cinemas after major chains declined to screen it; corporate-secret exposure including unreleased films; estimated tens of millions of dollars in direct response costs; significant international political consequences following US sanctions against North Korea
- **Investigators:** FBI (attributed attack to North Korea on 19 December 2014), FireEye / Mandiant for incident response, US intelligence community; independent skepticism expressed by Wired, the LA Times and by former hacker Hector Monsegur
- **Remediation:** Network rebuild; employee data-protection support; FBI investigation; sanctions imposed against North Korean entities and individuals in January 2015

Disinformation Details

- **Nature:** Textbook hybrid cyber-influence operation — combining hack-and-leak, hack-and-destroy (wiper), and coercive demands aimed at altering corporate behaviour; with a false-flag persona masking state attribution

- **Narratives:** (1) "Guardians of Peace" framed as an independent hacktivist group seeking justice rather than as a state-backed operation; (2) demands for Sony to "pull" *The Interview*; (3) 9/11-style threats of violence against cinemas screening the film; (4) public release of executive correspondence framed as exposing Sony's misconduct (off-colour remarks about Angelina Jolie, Aaron Sorkin and President Obama featured prominently in media coverage)
- **Deception:** (1) "Guardians of Peace" false-flag persona; (2) staged sequential leaks engineered to sustain media attention over weeks; (3) selectively framed disclosures designed to embarrass specific executives and accelerate internal disruption; (4) deniability of state involvement
- **Platforms:** Pastebin and other file-sharing services for data drops; mainstream press amplification of leaked emails; direct threats sent to Sony and to news outlets; cinema-chain communications
- **Reach:** Global — saturated international news cycles for weeks, with executive emails covered extensively across mainstream press
- **Impact:** Sony withdrew the wide theatrical release of *The Interview*; senior executives departed; Western corporations broadly reassessed their exposure to nation-state hack-and-leak operations; the incident became a foundational case study in cyber-enabled coercive influence
- **Disinfo Investigators:** FBI, NSA, FireEye / Mandiant, Wired's Kim Zetter, Hollywood Reporter, Trend Micro and many subsequent academic analyses
- **Remediation:** US sanctions on North Korea (January 2015); industry-wide reforms in protective-monitoring and breach-response; new awareness of executive-email exposure risk
- **Link to Cyberattack:** Every influence element depended on the underlying intrusion — leaked emails, wiped systems, and credible threats of further disclosure formed the coercive lever

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NA-14

MGM Resorts Scattered Spider vishing and ALPHV ransomware attack

% fit: 1.0

Date	Initial intrusion 7-8 September 2023; public disclosure 11 September 2023; ransomware deployment 11-17 September 2023
Targets	MGM Resorts International and its more than 30 properties globally, including the MGM Grand, Bellagio, Aria and Cosmopolitan in Las Vegas; subsequent customer-data disclosure affected MGM customers who had used MGM services before March 2019; Caesars Entertainment was hit by the same actors approximately one week earlier
Continent	North America
Perpetrator	Scattered Spider (tracked as UNC3944, a financially motivated English-speaking group with members in the US and UK known for advanced social-engineering) operating as an affiliate of the ALPHV / BlackCat ransomware-as-a-service operation
Motivation	Financial extortion through ransomware-as-a-service, exploiting MGM's high revenue, public profile, and reliance on networked operational systems including slot machines, digital room keys and reservations

Cyberattack Details

- **Nature:** Voice-phishing (vishing) of the IT help desk, abuse of identity-and-access management (Okta) for privilege escalation, lateral movement, and finally ALPHV / BlackCat ransomware deployment
- **Systems Attacked:** MGM corporate IT, Okta single-sign-on infrastructure, internal networks, slot-machine networks, digital room-key systems, online reservation systems, email and mobile-services infrastructure
- **Data Held:** Personal customer contact details and government-ID information for customers who had used MGM services before March 2019
- **Availability Impact:** Severe — widespread outages across MGM's Las Vegas properties; ATMs, slot machines, digital room keys, electronic payment systems, TV services and phone lines were affected; staff fell back to pen-and-paper processing for days
- **Data Exfiltration/Alteration:** Customer PII exfiltrated; ALPHV operators threatened to notify HaveIBeenPwned if negotiations failed
- **Estimated Impact:** Approximately US\$100 million in lost revenue plus legal, consulting and remediation costs; daily revenue loss of approximately US\$8.4 million during the outage; multiple class-action lawsuits; long-term reputational damage; Caesars Entertainment, hit by the same actors a week earlier, reportedly paid an undisclosed ransom (estimated \$15M)
- **Investigators:** MGM internal security and IR teams, Mandiant and other external responders, FBI, the vx-underground research community which obtained early threat-actor statements, Okta (which confirmed five clients including MGM were affected)
- **Remediation:** MGM refused to pay the ransom; phased system restoration; significant overhaul of help-desk authentication processes; class-action settlements; industry-wide reassessment of help-desk verification across the hospitality and casino sectors

Disinformation Details

- **Nature:** Pretext-based social-engineering / impersonation — the influence component was the convincing identity-claim made to the help desk
- **Narratives:** "I am an MGM employee and I need a credential reset"; subsequent ALPHV public statements positioning themselves as professional and ethical operators, including a public "Setting the record straight" statement

- **Deception:** (1) LinkedIn-based reconnaissance to identify a target employee; (2) vishing the IT help desk while impersonating that employee, leveraging English fluency to overcome cultural defences typical of Eastern-European-aligned ransomware actors; (3) ALPHV public statements designed to manage media narrative and pressure MGM into negotiation
- **Platforms:** Telephone (vishing), LinkedIn (reconnaissance), ALPHV public statement on the group's dark-web leak site, vx-underground's social-media platform, mainstream press coverage
- **Reach:** Global business and cybersecurity press; case-study status across the industry
- **Impact:** Catalysed industry-wide rethinking of help-desk authentication; established Scattered Spider as a leading social-engineering threat; demonstrated that vishing of a single employee can defeat enterprise-grade technical controls including MFA and Okta
- **Disinfo Investigators:** vx-underground, BleepingComputer, Krebs on Security, Netwrix, Specops, Cymulate, Reuters, NBC and major financial press
- **Remediation:** MGM internal reforms; sector-wide help-desk authentication hardening; FBI advisories and CISA alerts on Scattered Spider TTPs
- **Link to Cyberattack:** The entire initial-access vector was a social-engineering / impersonation operation — the influence component made the cyber operation possible

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NA-15

JBS Foods REvil ransomware attack on global meat-processing operations

% fit: 1.0

Date	30 May - 9 June 2021 (ransom paid on or around 9 June 2021; FBI public attribution to REvil on 2 June 2021)
Targets	JBS S.A., Brazil-headquartered, world's largest meat processor (approximately one-fifth of global meat supply by sales); meat-processing facilities in the United States, Canada and Australia; downstream global meat-supply chain and consumers
Continent	North America (the JBS USA subsidiary in Greeley, Colorado was the primary point of impact; JBS S.A. is Brazil-headquartered but Brazilian operations were not the focal point of the attack)
Perpetrator	REvil (also known as Sodinokibi), a Russian-speaking ransomware-as-a-service gang publicly attributed by the FBI on 2 June 2021
Motivation	Financial extortion through ransomware-as-a-service; REvil had publicly stated in October 2020 that the agricultural sector would become a primary target, reflecting opportunistic targeting of high-impact critical-supply-chain operators

Cyberattack Details

- **Nature:** Ransomware deployment with double-extortion mechanics (data exfiltration plus encryption) against JBS USA's IT systems
- **Systems Attacked:** JBS USA's North American and Australian IT infrastructure; the attack disabled beef and pork slaughterhouses across the company's facilities
- **Data Held:** Not extensively disclosed publicly; the focus of the attack was operational disruption rather than highly sensitive personal data
- **Availability Impact:** Severe — JBS shut down all of its North American and Australian beef plants temporarily; pork, poultry and prepared-foods plants were also disrupted; Australian operations were idled for days; recovery was staged over approximately a week
- **Data Exfiltration/Alteration:** Data exfiltration was implied by the ransom payment but precise volumes were not publicly disclosed
- **Estimated Impact:** US\$11 million ransom paid in Bitcoin (one of the largest publicly confirmed ransomware payments at the time); short-term global meat-supply concerns, including spot disruption of beef and pork wholesale markets; significant regulatory and political reaction, including US-Russia diplomatic engagement; recognition of food-sector as critical infrastructure deserving cyber-defence priority
- **Investigators:** FBI (issued public attribution to REvil on 2 June 2021), CISA, US Department of Agriculture, Australian and Canadian government partners; JBS internal teams supported by external IR firms
- **Remediation:** JBS confirmed payment of US\$11 million ransom on 9 June 2021 to prevent further disruption; phased operational restart; investment of more than US\$200 million annually in IT and 850+ IT professionals globally cited by JBS as ongoing baseline; broader policy consequences include direct Biden-Putin discussion on 9 July 2021 about ransomware operations from Russian soil

Disinformation Details

- **Nature:** Standard ransomware double-extortion coercion — pay or face publication of stolen data and continued operational disruption; the REvil operation was professionalised but not ideologically branded
- **Narratives:** Implicit pay-or-publish coercion; REvil maintained a public leak-site presence but did not post JBS data, consistent with payment having been made

- **Deception:** Standard ransomware operational security; REvil operating through Tor-hosted negotiation portals and pseudo-professional "help desk" operations
- **Platforms:** REvil's dark-web leak site, ransom-negotiation portals, downstream global press coverage including NPR, Reuters, Bloomberg, Al Jazeera, BBC
- **Reach:** Global — JBS supplies roughly 20% of global meat by sales, so the attack received saturation news coverage in dozens of countries
- **Impact:** Public confirmation by CEO Andre Nogueira that JBS paid US\$11 million to "prevent any potential risk for our customers"; significant policy debate over the wisdom of paying; political pressure on Russia regarding harbouring of ransomware gangs; subsequent US-Russia diplomatic engagement on cybercrime
- **Disinfo Investigators:** FBI, Recorded Future, journalists at NPR / Al Jazeera / Wikipedia community / Reuters / Bloomberg; ongoing critical academic and policy analysis of agriculture-sector ransomware
- **Remediation:** Public disclosure by JBS; broader sectoral attention to food-system cyber resilience; eventual REvil disruption operations by international partners later in 2021
- **Link to Cyberattack:** Any extortion pressure depended entirely on the underlying ransomware deployment and on credible threats of further disruption and data exposure

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NA-16

Change Healthcare ALPHV / BlackCat ransomware, exit-scam and RansomHub re-extortion

% fit: 1.0

Date	Initial intrusion 11-12 February 2024; nine-day lateral movement; ransomware deployed 21 February 2024; US\$22 million ransom paid early March 2024; ALPHV exit-scam and RansomHub re-extortion through April-May 2024
Targets	Change Healthcare (a UnitedHealth Group subsidiary processing approximately half of all US medical claims for around 900,000 physicians, 33,000 pharmacies, 5,500 hospitals and 600 laboratories); ultimately compromised the personal, health, and financial information of approximately 100 million Americans
Continent	North America
Perpetrator	ALPHV / BlackCat ransomware-as-a-service operation; an affiliate of the group performed the actual intrusion and then was reportedly cheated by ALPHV's "exit scam"; the affiliate subsequently took the stolen data to a second ransomware group, RansomHub, for further extortion
Motivation	Financial extortion through ransomware, with double-extortion mechanics; opportunistic targeting of a high-value healthcare-data processor with critical national role

Cyberattack Details

- **Nature:** Credential-based intrusion into an unprotected Citrix remote-access portal lacking multi-factor authentication, followed by nine days of lateral movement, then ALPHV / BlackCat ransomware deployment; further compounded by post-payment RansomHub re-extortion
- **Systems Attacked:** Change Healthcare's nationwide medical-claims, billing and prescription-processing systems, plus broader UnitedHealth Group infrastructure (UnitedHealth severed connectivity to Change's data center to prevent malware spread to wider corporate systems)
- **Data Held:** Personal, health, and financial information of approximately 100 million individuals — names, contact details, Social Security numbers, driver's licence numbers, health information, insurance information, billing information, patient diagnoses, treatment information and health-insurance member IDs
- **Availability Impact:** Severe — weeks-long disruption of pharmacy operations, medical-claims processing, billing and prescription services across the US; patients forced to pay out-of-pocket for medications; thousands of physician practices, pharmacies and small healthcare providers faced cash-flow crises
- **Data Exfiltration/Alteration:** Approximately 6TB of data exfiltrated and held first by the ALPHV affiliate, then by RansomHub after the affiliate moved to that group following the exit-scam
- **Estimated Impact:** Largest healthcare data breach in US history; approximately US\$2.87 billion in response costs incurred by UnitedHealth Group in 2024; over US\$6 billion in financial assistance provided to affected healthcare providers; US\$22 million ransom paid to ALPHV; second extortion demand from RansomHub (not paid); major reputational and political consequences, including testimony from UnitedHealth CEO Andrew Witty before Congress on 1 May 2024 confirming the ransom payment
- **Investigators:** Mandiant (incident response), FBI, CISA, US Department of Health and Human Services Office for Civil Rights (HIPAA breach investigation); subsequent state-level investigations including Nebraska Attorney-General lawsuit
- **Remediation:** US\$22 million ransom paid by UnitedHealth Group to ALPHV with the goal of preventing data publication; ALPHV exit-scammed its affiliate, who retained the data; affiliate took stolen data to RansomHub which made a second extortion demand (not paid); UnitedHealth provided large-scale provider financial assistance; congressional hearings; multi-state class-action and AG litigation; HHS-OCR HIPAA enforcement

Disinformation Details

- **Nature:** Multi-layered extortion-and-coercion campaign with substantial reputational and public-narrative dimensions; the ALPHV exit-scam and RansomHub re-extortion produced an unusually public threat-actor-versus-threat-actor disinformation episode
- **Narratives:** (1) ALPHV's pay-or-publish coercion against UnitedHealth; (2) ALPHV's post-payment public statement and shutdown narrative designed to disguise an exit-scam against its own affiliate; (3) RansomHub's framing of itself as legitimate inheritor of the stolen data and as authorised re-extortionist; (4) public-narrative pressure on UnitedHealth via congressional and media accountability for the payment decision
- **Deception:** ALPHV's fake "law enforcement seizure" claims on its own leak site, designed to convince its affiliate community that the operation had been disrupted and to hide the exit-scam; RansomHub's claim of legitimate authority to extort with the stolen data
- **Platforms:** ALPHV and RansomHub dark-web leak sites, US Department of Justice / FBI public statements, congressional testimony (Senate Finance Committee and House Energy and Commerce subcommittee), mainstream press coverage (Krebs on Security, IBM, BlackFog, HIPAA Journal)
- **Reach:** Massive — directly affected approximately 100 million Americans and a substantial portion of the US healthcare-payment system; saturated US news coverage for months
- **Impact:** Public discrediting of ransomware-payment-as-protection model; explicit congressional testimony admitting the futility of the payment ("did not get its data back"); broader policy debate about banning ransom payments; ALPHV ceased operations, RansomHub emerged as a major new player partly absorbing ALPHV affiliates
- **Disinfo Investigators:** Krebs on Security, IBM Security, BlackFog, HIPAA Journal, Holland & Knight legal analysis, Aviatix, Hyperproof, congressional staff and law-enforcement; Nebraska AG litigation discovery
- **Remediation:** Public disclosure by UnitedHealth CEO; congressional accountability; HHS-OCR HIPAA enforcement; broader US healthcare-sector security reforms
- **Link to Cyberattack:** Every influence and coercion element depended on the underlying intrusion and exfiltration; this incident is a particularly stark example of how ransomware extortion campaigns are simultaneously cyber attacks and influence operations

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Africa

This table classifies 20 incidents targeting African entities according to the cyber–influence taxonomy:

- **(1) Cyber-enabled influence** — cyber operations serving an influence goal (1a–1h)
- **(2) Influence-enabled cyber** — disinformation/deception enabling a cyber intrusion (2a–2h)
- **(3) Parallel cyber–influence** — simultaneous, independent cyber and influence operations by the same actor

Where an incident is predominantly a coordinated-inauthentic-behaviour (CIB) campaign built on cyber-enabled fake-account / fake-site infrastructure but does not map cleanly to a numbered sub-type, it is classified under **1c (broad reading: injecting propaganda via cyber-enabled fake assets and impersonation infrastructure)**, which best reflects the taxonomy's framing of "injecting propaganda or disinformation into websites".

- **1.0** — textbook fit (defining act of the sub-type clearly present).
- **0.7–0.9** — strong fit, partial or mixed on one element.
- **≤ 0.5** — capped here for Type-1 incidents without a real unauthorised intrusion or DDoS, and for Type-2 incidents lacking a tactically-connected subsequent intrusion or DDoS by the same actor.
- **Type-3** scored on whether the cyber and influence components are by the same threat actor and close in time.

Incidents included (% fit ≥ 0.7): 9 of 20

ID	Incident	Country/Scope	Primary Type	Secondary Type(s)	Perpetrator	Victim Category	% Fit
AF-6	Industrial Ransomware and Data Extortion Against South African Organizations	South Africa	2f – doxxing-based (double-extortion) ransomware	1b – hack-and-leak (samples posted to leak sites for coercive shaming)	Cybercriminal ransomware groups (BianLian, RansomHub, MeowLeaks, BlackSuit, MedusaLocker, Cactus) — mixed/uncertain	Private sector / critical infrastructure (industrial organisations)	1.0
AF-9	RansomEXX Extorts Kenya Airways With Leak-Site Doxing Threats	Kenya	2f – doxxing-based (double-extortion) ransomware	1b – hack-and-leak (selective publication of stolen records to pressure target)	Cybercriminal group RansomEXX — mixed/uncertain	Private sector (aviation), passengers, regulators	1.0
AF-14	Transnet ransomware and port force-majeure disruption	South Africa	2f – doxxing-based (double-extortion) ransomware	1d – hack-and-destroy (operational shutdown of port and rail IT, force majeure declared)	Cybercriminal group deploying "Death Kitty" / HelloKitty ransomware (Russia / Eastern Europe attribution per CrowdStrike)	Private sector / critical infrastructure (ports, rail, supply chain)	0.9
AF-15	TransUnion South Africa breach and N4ughtySecTU "insurance fee" extortion	South Africa	2f – doxxing-based (double-extortion) ransomware (data-theft variant)	1b – hack-and-leak (threatened publication of stolen credit-bureau and client data)	N4ughtySecTU / N4ughtySecGroup (self-identified Brazilian extortion crew)	Private sector (credit bureau, downstream banks), consumers	1.0
AF-16	Experian South Africa breach via fraudulent-client impersonation	South Africa	2d – website / account impersonation (fraudster posed as a legitimate Experian client)	2g – false-flag operation (false legitimate-client identity claim)	Karabo Phungula (South African fraudster, later convicted)	Private sector (credit bureau), consumers, businesses	0.9
AF-17	Cell C RansomHouse double-extortion and customer-data leak	South Africa	2f – doxxing-based (double-extortion) ransomware	1b – hack-and-leak (publication of stolen customer records on dark-web leak site)	RansomHouse data-extortion group	Private sector (telecoms), subscribers, employees	1.0
AF-18	Sibanye-Stillwater global IT cyberattack on African mining group	South Africa (and global operations)	2f – doxxing-based (double-extortion) ransomware	1d – hack-and-destroy (global IT outage, manual fallback processes)	Unattributed (Sibanye declined to engage with attackers)	Private sector (mining), employees, shareholders	0.8

ID	Incident	Country/Scope	Primary Type	Secondary Type(s)	Perpetrator	Victim Category	% Fit
AF-19	Astral Foods cyberattack and poultry-supply disruption	South Africa	2f – doxxing-based (double-extortion) ransomware	1d – hack-and-destroy (poultry-division processing and delivery downtime)	Unattributed (no ransomware group publicly claimed credit)	Private sector (agriculture / poultry / animal feed)	0.7
AF-20	MTN Group cybersecurity incident and pan-African subscriber-data breach	South Africa (HQ) and 18 African / Middle Eastern markets	2f – doxxing-based (double-extortion) ransomware (data-theft variant)	1b – hack-and-leak (threatened release of subscriber data)	Unattributed ("unknown third party" per MTN)	Private sector (telecoms), subscribers	0.8

Continental observations

- **Dominant pattern across all 20 incidents:** a clear split between cyber-enabled influence via CIB infrastructure (Type 1c, predominantly Russian-linked or domestic political-PR networks) and double-extortion ransomware against private-sector targets (Type 2f, with leak-site doxxing as the coercive lever).
- **Civil-society / electoral incidents (1c):** 11 cases, primarily attributable to Russian-linked operators (Prigozhin-era / Wagner-aligned networks) and to domestic political-PR networks.
- **Corporate / private-sector incidents (predominantly 2f):** 9 cases — including ports and rail (Transnet), credit bureaus (TransUnion, Experian), telecoms (Cell C, MTN), mining (Sibanye-Stillwater), poultry (Astral Foods), aviation (Kenya Airways), and a wave of industrial cases (AF-6). All but the Experian case follow the ransomware double-extortion playbook; Experian is an exemplary case of pure influence-enabled cyber via legitimate-client impersonation.
- **Most-affected victim categories:** elections and voters, civil society, and a rapidly growing tail of private-sector / critical-infrastructure targets across banking, telecoms, mining, agriculture and logistics.
- **South Africa-specific observation:** the country accounts for nine of the 20 African cases and is the centre of the continent's emerging double-extortion-ransomware epidemic.

AF-6

Industrial Ransomware and Data Extortion Against South African Organizations

% fit: 1.0

Date	Q4 2024
Targets	Industrial and infrastructure-linked organizations in South Africa, along with their employees, customers, and partners whose data or operations could be used for extortion leverage
Continent	Africa
Perpetrator	Multiple ransomware and extortion actors, with reporting highlighting BianLian, RansomHub, MeowLeaks, BlackSuit, MedusaLocker, and Cactus in the broader industrial threat ecosystem
Motivation	Extort payment through stolen-data exposure, leak-site shaming, and in some cases operational disruption against industrial victims in a relatively underprotected regional market

Cyberattack Details

- **Nature:** Double-extortion ransomware and data-extortion campaigns targeting industrial organizations
- **Systems Attacked:** Industrial enterprise networks, VPN appliances, remote desktop services, and business systems connected to industrial operations
- **Data Held:** Exfiltrated business data, sensitive internal records, and operationally relevant information used to pressure victims through leak-site threats
- **Availability Impact:** Some industrial ransomware campaigns globally caused operational disruption, while others shifted toward data-extortion-heavy models without emphasizing encryption
- **Data Exfiltration/Alteration:** Threat reporting indicates large-scale exfiltration used as leverage, especially as BianLian evolved toward data-extortion-centric tactics and collaborated with RansomHub tooling and infrastructure
- **Estimated Impact:** Africa saw seven reported industrial ransomware incidents in Q4 2024, with South Africa and Tunisia accounting for the most reported attacks in the region
- **Investigators:** Dragos and other industrial-threat analysts
- **Remediation:** Threat-intelligence reporting, monitoring of ransomware leak sites, and hardening of exposed remote access infrastructure

Disinformation Details

- **Nature:** Public leak-site coercion and reputational pressure rather than a classic ideological disinformation campaign
- **Narratives:** Victims are vulnerable, data will be exposed, and failure to pay will result in escalating public embarrassment or downstream business harm
- **Deception:** Ransomware groups routinely use curated leak posts, countdown pressure, and selective disclosure of stolen records to magnify perceived consequences and accelerate payment decisions
- **Platforms:** Criminal leak sites and extortion portals
- **Reach:** Primarily targeted at executives, customers, regulators, insurers, and partners who may react to publicized compromise or data loss
- **Impact:** Even where operational disruption is limited, the doxing threat transforms the intrusion into a psychological pressure campaign aimed at forcing negotiation and payment
- **Disinfo Investigators:** Industrial-threat researchers and ransomware tracking services
- **Remediation:** Network hardening, leak-site monitoring, and incident-response preparations tailored to extortion and reputational fallout

- **Link to Cyberattack:** The coercive messaging depends on data stolen through network intrusion and is designed to increase the extortion value of the original attack

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AF-9

RansomEXX Extorts Kenya Airways With Leak-Site Doxing Threats

% fit: 1.0

Date	December 2023-January 2024
Targets	Kenya Airways, its passengers, staff, associated government officials, and the airline's public reputation
Continent	Africa
Perpetrator	RansomEXX ransomware group
Motivation	Extort payment by threatening to publish and publicize stolen sensitive data, while increasing reputational, regulatory, and operational pressure on a flagship African airline

Cyberattack Details

- **Nature:** Ransomware-linked intrusion with data theft and leak-site extortion
- **Systems Attacked:** Kenya Airways information systems and associated data repositories containing passenger, staff, and internal corporate records
- **Data Held:** Contact details, identification documents, passport copies, police and investigation reports, accident records, staff-related records, and internal corporate or planning documents
- **Availability Impact:** Kenya Airways indicated operations were not significantly impacted, suggesting the main pressure mechanism was data theft and exposure rather than severe service disruption
- **Data Exfiltration/Alteration:** Sensitive data was exfiltrated and samples were posted on the group's leak site, with more disclosure threatened if ransom demands were not met
- **Estimated Impact:** Privacy risk to passengers and staff, reputational damage to a national carrier, exposure of sensitive investigative and official records, and coercive pressure through public leak-site shaming
- **Investigators:** Kenya Airways sources cited in press reporting, leak-site monitoring services, and cybersecurity reporting outlets
- **Remediation:** Kenya Airways declined the ransom demand; public reporting and leak-site monitoring exposed the scope of the incident

Disinformation Details

- **Nature:** Coercive public doxing threat used as an extortion and pressure tactic rather than a broader political disinformation campaign
- **Narratives:** The victim cannot protect sensitive passenger and official information; more damaging disclosures will follow unless payment is made
- **Deception:** Leak-site posts selectively showcased sensitive material to maximize fear, embarrassment, and pressure on the airline and stakeholders
- **Platforms:** Dark-web leak site and downstream media coverage of the breach
- **Reach:** Public enough to affect customer trust, regulatory scrutiny, and external perceptions of the airline's security posture
- **Impact:** The threat of disclosure turned the cyber intrusion into a reputational and psychological pressure campaign aimed at forcing payment and undermining stakeholder confidence
- **Disinfo Investigators:** Journalists and breach-monitoring services documenting the leak-site disclosures
- **Remediation:** Refusal to pay, incident disclosure, and continued monitoring of leak-site releases

- **Link to Cyberattack:** The pressure campaign relied on stolen data from the intrusion and the threat of wider publication as the core extortion mechanism

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AF-14

Transnet ransomware attack and force-majeure disruption of South African ports

% fit: 0.9

Date	22 July 2021
Targets	Transnet SOC Ltd; container terminals at Durban, Cape Town, Port Elizabeth and Ngqura; South African and regional supply chains for mining, agricultural and manufactured exports
Continent	Africa
Perpetrator	Threat actor deploying ransomware linked by CrowdStrike to the "Death Kitty" / "HelloKitty" / "Five Hands" family, assessed by researchers to originate in Russia or Eastern Europe
Motivation	Financial extortion through encryption and threatened publication of stolen data, against a high-value state-owned operator of critical maritime and rail infrastructure

Cyberattack Details

- **Nature:** Network intrusion, file encryption, and double-extortion ransomware deployment with a 1TB exfiltration claim
- **Systems Attacked:** Transnet's IT environment, including the Navis container operating system used at port terminals; corporate, port, rail, pipeline and engineering systems were variously affected
- **Data Held:** Ransom note claimed exfiltration of approximately 1TB of personal data, financial reports and other internal documents
- **Availability Impact:** Severe — Transnet declared force majeure across container terminals and switched to a paper-based manual clearance process; Port of Durban was reported operating at 10% capacity at the height of the disruption, with truck turnaround times exceeding 14 hours
- **Data Exfiltration/Alteration:** Attackers claimed encryption of files and exfiltration of 1TB; Transnet later stated no raw customer data had been compromised
- **Estimated Impact:** Disruption of more than 60% of South Africa's container traffic for over a week; regional knock-on effects for landlocked neighbours and for SA's mining and agricultural exports; force-majeure declaration limiting Transnet's contractual liability
- **Investigators:** Transnet incident-response team, South African government (Department of Public Enterprises), CrowdStrike (attribution analysis), Bloomberg and ISS Africa reporting
- **Remediation:** No ransom paid; Active Directory rebuild, Microsoft E5 stack deployment fast-tracked, OS upgrades and patching, deployment of WAF / reverse-proxy / anti-DDoS infrastructure for public sites; ~90% of IT systems restored within two weeks

Disinformation Details

- **Nature:** Coercive leak-site / ransom-note extortion rather than ideological influence operation
- **Narratives:** Pay or stolen records will be released; the company's critical infrastructure cannot be defended without negotiation
- **Deception:** Use of dark-web negotiation portal, ambiguity around volume and sensitivity of exfiltrated data, and reliance on operational fear (force-majeure conditions) to pressure payment
- **Platforms:** Ransom note on victim systems, dark-web negotiation portal, downstream press coverage
- **Reach:** National and international through extensive South African and global press coverage of the force-majeure event

- **Impact:** Even without payment, the public disruption to a flagship state-owned enterprise had reputational and political fallout; the attack came days after the 2021 KwaZulu-Natal unrest, fuelling unverified public speculation of coordinated economic destabilisation (authorities concluded the two events were unrelated)
- **Disinfo Investigators:** Public-sector spokespersons, journalists at Bloomberg / Reuters / Daily Maverick / TechCentral, and ISS Africa analysts
- **Remediation:** Public attribution analysis by CrowdStrike; government communications; ongoing parliamentary and regulatory scrutiny
- **Link to Cyberattack:** The coercive pressure depended entirely on the encryption of operational systems and the threat of releasing exfiltrated data

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AF-15

TransUnion South Africa breach and N4ughtySecTU "insurance fee" extortion campaign

% fit: 1.0

Date	March 2022 (with renewed extortion demands in November 2023 and further claims in 2024)
Targets	TransUnion South Africa, its corporate clients (Absa, FNB, Standard Bank, Nedbank, Discovery, TymeBank and others), and millions of South African consumers whose credit data was hosted on the breached server
Continent	Africa
Perpetrator	N4ughtySecTU (later N4ughtySecGroup), a self-identified Brazilian extortion crew
Motivation	Financial extortion of TransUnion through ransom demand, plus secondary extortion of TransUnion's downstream corporate clients via threatened release of their customer records

Cyberattack Details

- **Nature:** Brute-force attack against an internet-facing SFTP server using stolen / weak client credentials reportedly protected by the password "Password", leading to bulk data exfiltration; no ransomware encryption
- **Systems Attacked:** A TransUnion South Africa server holding credit-bureau data, accessed via a compromised authorised-client account
- **Data Held:** Names, ID numbers, dates of birth, gender, contact details, marital status, financial and credit-related records; attackers claimed access to records of 54 million individuals and approximately 200 corporate clients
- **Availability Impact:** Limited — TransUnion took some services offline as a precaution but credit-bureau core operations resumed within days
- **Data Exfiltration/Alteration:** Up to 4TB of records claimed by the attackers; TransUnion later confirmed at least 3 million South African consumers were directly impacted, with approximately 6 million additional ID numbers identified
- **Estimated Impact:** Loss of sensitive credit-history data for millions of South Africans; secondary exposure of customer datasets used by major SA banks; substantial regulatory and reputational fallout; renewed extortion of \$30M each from TransUnion and Experian in November 2023
- **Investigators:** TransUnion incident-response team, external cybersecurity / digital-forensic experts, South African Information Regulator and law enforcement
- **Remediation:** TransUnion publicly refused to pay; free identity-protection products offered to affected consumers; access by the compromised client was suspended

Disinformation Details

- **Nature:** Double-extortion-style coercion combined with a novel "insurance fee" campaign — TransUnion's corporate clients were approached separately and asked to pay smaller "insurance" sums (~\$1M for large clients, \$100K for smaller ones) to be excluded from any future leak
- **Narratives:** TransUnion's data is in the attackers' hands; downstream institutions cannot protect their customers' records; payment is the only protection
- **Deception:** Claims of continued persistent access through 2023 and 2024 (disputed by TransUnion and Experian), use of multiple Telegram and Session channels for negotiation, and weaponised lists of allegedly affected client institutions
- **Platforms:** ITWeb / MyBroadband / TimesLive / DataBreaches.net press coverage; Telegram and Session for direct contact with victims

- **Reach:** National (every adult South African with a credit record is plausibly affected) and international through credit-bureau interconnections
- **Impact:** Sustained reputational pressure on TransUnion, Experian and the South African financial-services sector; public anxiety driven by claims of ongoing access; political pressure on the Information Regulator
- **Disinfo Investigators:** MyBroadband, ITWeb, BleepingComputer, BitDefender, Stellenbosch University researchers Joel Cedras and Veer Gosai (separate but related SASSA-fraud disclosures)
- **Remediation:** Public refusal to pay; ongoing Information Regulator engagement; consumer protective-registration services via SAFPS
- **Link to Cyberattack:** The "insurance fee" influence layer depended directly on the original intrusion and on credible claims of held data

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AF-16

Experian South Africa breach via fraudulent-client impersonation

% fit: 0.9

Date	August 2020
Targets	Experian South Africa and the personal records of approximately 24 million South African consumers and 793,749 business entities held in its credit-bureau database
Continent	Africa
Perpetrator	Karabo Phungula, a South African fraudster who posed as a legitimate Experian client to obtain bulk data; later convicted and sentenced to 15 years in 2023
Motivation	Acquisition of bulk consumer and business data for downstream fraud, identity-theft and marketing-lead resale, exploiting Experian's client-onboarding and data-release processes

Cyberattack Details

- **Nature:** Sophisticated social-engineering / impersonation attack — no malware deployment, no system intrusion in the technical sense; the perpetrator successfully impersonated a representative of a legitimate Experian client and requested data through standard business channels
- **Systems Attacked:** Experian South Africa's client-onboarding, identity-verification and bulk-data-release workflows
- **Data Held:** Names, ID numbers, contact details, employment information, and limited business records for ~24 million individuals and ~793,749 businesses
- **Availability Impact:** None — operations continued normally; this was a confidentiality breach via authorised channels
- **Data Exfiltration/Alteration:** Bulk consumer and business records were handed over by Experian to the fraudster on the basis of false client identity; no unauthorised system access required
- **Estimated Impact:** Largest single data exposure in South African history at the time; substantial downstream identity-theft and phishing risk; the South African Banking Risk Centre (SABRIC) flagged the incident as a systemic threat to financial-sector fraud controls
- **Investigators:** Experian, SABRIC, South African Information Regulator, South African Police Service; eventual prosecution by the Specialised Commercial Crimes Court
- **Remediation:** Experian asserted that recovered data was secured before wider distribution; ongoing regulatory engagement; conviction of Phungula in March 2023 with a 15-year sentence

Disinformation Details

- **Nature:** The deception was the entire intrusion vector — there was no separate "FIMI" campaign, but the case is a textbook influence-enabled cyber incident in which fraudulent identity claims achieved what conventional hacking could not
- **Narratives:** "I am a legitimate Experian client and I need this data for lawful marketing-lead generation"
- **Deception:** Pretexting and identity impersonation directed at Experian's business processes, including credible mimicking of an authorised commercial client
- **Platforms:** Experian's business-account onboarding and customer-service channels; no public-facing influence component
- **Reach:** Direct (one perpetrator, one target firm) but with systemic consequences for 24 million data subjects
- **Impact:** Demonstrated that bulk credit-bureau data could be exfiltrated without any technical exploit, simply by exploiting weak client-identity-verification controls — a finding with continental implications for other African credit bureaus, telecoms and financial-services providers

- **Disinfo Investigators:** Information Regulator, Herbert Smith Freehills legal analysis, SABRIC, journalists at TimesLive / TechTimes / News24
- **Remediation:** Reinforcement of Experian's client-vetting controls; broader sectoral push for stronger KYC at credit-bureau client onboarding
- **Link to Cyberattack:** The "cyber intrusion" in this case is the impersonation itself — the influence operation and the cyber operation are the same act

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AF-17

Cell C RansomHouse double-extortion and customer-data leak

% fit: 1.0

Date	Late 2024 (intrusion detected 20 December 2024; ransom demand 11 April 2024 per researchers; data published on dark web 28 December 2024; public confirmation by Cell C 8 January 2025; further confirmation of leak 9 April 2025)
Targets	Cell C, the third-largest mobile operator in South Africa, and approximately 7.7 million subscribers along with employees, partners and a former senior executive
Continent	Africa
Perpetrator	RansomHouse data-extortion group (active since late 2021 / early 2022; previous victims include AMD, Keralty, and South African retailer Shoprite)
Motivation	Financial extortion through threatened publication of stolen customer and corporate data, exploiting Cell C's role as a major SA telecoms provider holding rich identity and financial data

Cyberattack Details

- **Nature:** Multi-pronged data-extortion intrusion typical of RansomHouse — exfiltration-focused with no widespread encryption; reportedly preceded by phishing campaigns in 2023 according to researchers (disputed by Cell C)
- **Systems Attacked:** Unstructured-data repositories within Cell C's corporate IT environment; researcher analysis of leaked credentials suggested possible exposure of FTTH ordering, provisioning and billing portals linked to fibre operators (MetroFibre, Openserve, Vumatel)
- **Data Held:** Full names, contact details, email addresses, phone numbers, ID numbers, banking information, driver's licence numbers, medical records, passport details, and personal records of a former senior executive
- **Availability Impact:** Limited — Cell C network services and billing systems remained operational; the impact was predominantly on data confidentiality and corporate reputation
- **Data Exfiltration/Alteration:** Approximately 2TB of data exfiltrated and subsequently published on RansomHouse's dark-web leak site after Cell C did not engage with the ransom demand
- **Estimated Impact:** Significant identity-theft, SIM-swap and phishing risk for millions of Cell C subscribers; regulatory engagement with the South African Information Regulator; reputational damage during a sensitive period of corporate restructuring
- **Investigators:** Cell C's external cybersecurity advisers, threat-intelligence firm TFI (which mapped leaked data and exfiltration timeline), SentinelOne analysis of RansomHouse tradecraft, Information Regulator and SAPS
- **Remediation:** Containment and isolation of affected systems; engagement of external cybersecurity experts; public information-hub launched to advise stakeholders on protective registration via SAFPS; no ransom paid

Disinformation Details

- **Nature:** Coercive leak-site doxing campaign — RansomHouse markets itself publicly as a "force for good" exposing corporate security failings, a posture designed to add reputational pressure beyond conventional ransom demands
- **Narratives:** Cell C cannot protect customer data; the company concealed the incident's true scope; releasing data is "ethical disclosure" rather than crime
- **Deception:** Selective publication of records on the leak site; messaging framing exfiltration as a public-interest act; partial disclosure designed to maximise media uptake
- **Platforms:** RansomHouse dark-web leak site; downstream coverage by Connecting Africa, TechCentral, BleepingComputer, Security Affairs and The Record
- **Reach:** National (SA mass-market subscribers) and international (telecoms sector and threat-intelligence community)

- **Impact:** Sustained negative press coverage during a delicate period for Cell C's corporate finances; renewed attention to ransomware risks in African telecoms; consumer alarm despite Cell C's assertion that core billing and authentication systems were not breached
- **Disinfo Investigators:** TFI, SentinelOne, FalconFeeds.io, Connecting Africa, TechCentral and other sector press
- **Remediation:** Public statements correcting RansomHouse's narrative framing; investor and regulator communications; information-hub for affected stakeholders
- **Link to Cyberattack:** The reputational pressure depended entirely on the exfiltrated data; the influence component is wholly cyber-enabled

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AF-18

Sibanye-Stillwater global IT cyberattack affecting a major African mining group

% fit: 0.8

Date	Intrusion mid-June 2024; discovered 8 July 2024; publicly disclosed 11 July 2024; data-breach notifications to affected employees in mid-August 2024
Targets	Sibanye-Stillwater, a JSE- and NYSE-listed multinational mining and metals-processing group with operations across South Africa, the United States (Montana), Zimbabwe, the Democratic Republic of Congo and Argentina; over 7,000 employees and their personal records were ultimately affected
Continent	Africa
Perpetrator	Not publicly attributed; Sibanye-Stillwater chose not to engage with the attackers and did not disclose the threat-actor identity or ransom demand
Motivation	Financial extortion through ransomware against a high-value mining target with sensitive personnel, payroll, customer and supplier data and significant share-price exposure

Cyberattack Details

- **Nature:** Ransomware-class intrusion with data exfiltration; precise variant not publicly disclosed
- **Systems Attacked:** Sibanye-Stillwater's global IT infrastructure, including systems at South African operations and US Montana subsidiaries; required temporary system outages and manual back-up processing
- **Data Held:** Employee personal information for at least 7,000 individuals (including names, SSN-equivalent identifiers, payroll-related data and other personally identifying information per the Montana class-action complaint)
- **Availability Impact:** Limited disruption to physical mining operations thanks to manual fallback processes, but the company invoked all disaster-recovery and preparedness protocols; the half-year 2024 interim financial results announcement was delayed by two weeks (rescheduled from 29 August to 12 September 2024)
- **Data Exfiltration/Alteration:** Confirmed exfiltration of employee personal data; full scope not publicly disclosed
- **Estimated Impact:** Financial reporting delay, regulatory engagement across five operating jurisdictions, class-action litigation in Montana on behalf of more than 7,000 employees, broad reputational impact on the African mining sector
- **Investigators:** Sibanye-Stillwater internal IT/security teams, external cybersecurity experts, multi-jurisdictional regulators
- **Remediation:** Phased system shutdown and controlled restart; hardening of IT security; explicit refusal to engage with attackers; voluntary disclosure to regulators in all affected countries

Disinformation Details

- **Nature:** Coercive ransom / leak-site extortion model — though Sibanye declined to disclose any specific extortion narrative or leak-site posting in public statements
- **Narratives:** Implicit — pay or face publication of stolen employee and corporate data; the explicit narrative was suppressed by the company's decision not to engage
- **Deception:** Standard ransomware-extortion psychological pressure, denied by Sibanye's refusal to negotiate
- **Platforms:** Internal ransom communications (not publicly disclosed); downstream press coverage in Moneyweb, ITWeb, Bizcommunity, News24, KTVQ Montana
- **Reach:** International — Sibanye-Stillwater is listed on the JSE and NYSE, with operations on five continents; news coverage was carried in South African, US and international mining press
- **Impact:** Possibility of corporate espionage and share-price manipulation flagged by ICT analysts; broader concerns about cyber risks to mining-sector automation and operational technology

- **Disinfo Investigators:** Sibanye-Stillwater corporate communications, sector analysts (Adrian Schofield and others), specialist mining and cybersecurity press
- **Remediation:** Refusal to engage; voluntary regulatory disclosure; class-action defence; tightening of mining-sector cyber posture across peer companies (Nampak, Wesizwe Platinum, Astral Foods also reported incidents in the same window)
- **Link to Cyberattack:** Any extortion pressure depended on the underlying cyber intrusion and on the threat to expose exfiltrated employee data

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AF-19

Astral Foods cyberattack and poultry-supply disruption

% fit: 0.7

Date	16 March 2025; publicly disclosed via voluntary trading update 24 March 2025
Targets	Astral Foods, South Africa's largest integrated poultry producer and largest animal-feed producer; subsidiaries including National Chicks, County Fair, Ross Poultry, Tiger Chicks, Tiger Animal Feeds and Provimi; downstream retail, wholesale and fast-food customers in South Africa and neighbouring countries
Continent	Africa
Perpetrator	Not publicly attributed; no ransomware group has claimed credit on a leak site as of disclosure
Motivation	Almost certainly financial extortion — incident bears the hallmarks of a ransomware attack, with operational disruption and recovery costs consistent with the standard double-extortion playbook

Cyberattack Details

- **Nature:** Cyber intrusion described by the company as a "cybersecurity incident" with all hallmarks of a ransomware attack; full disaster-recovery and preparedness protocols were activated
- **Systems Attacked:** Astral Foods' IT environment supporting poultry processing, deliveries and feed-manufacturing operations across the group's South African subsidiaries
- **Data Held:** Astral asserted that no confidential customer, supplier or stakeholder data was compromised; the precise scope and any exfiltration claim were not publicly disclosed
- **Availability Impact:** Significant — the Poultry Division experienced downtime in processing and deliveries, requiring manual back-up procedures and a production catch-up period
- **Data Exfiltration/Alteration:** Not confirmed publicly; the company's official line was that no sensitive data was compromised
- **Estimated Impact:** Approximately R20 million (~US\$1.1 million) profit hit during the reporting period due to lost revenue and production-backlog catch-up costs; contributed to a forecast half-year profit decline of up to 60% (compounded by lower chicken prices and constrained consumer spending in SA)
- **Investigators:** Astral Foods internal IT and incident-response teams, external cybersecurity advisers, JSE disclosure regime
- **Remediation:** Activation of disaster-recovery protocols; phased system restoration; voluntary investor trading update; no public confirmation of ransom payment or engagement with attackers

Disinformation Details

- **Nature:** Limited public disinformation footprint — no leak-site posting or extortion narrative was publicly evident at the time of disclosure
- **Narratives:** Implicit extortion pressure typical of ransomware operators, but not publicly visible; Astral's communications focused on operational recovery rather than threat-actor messaging
- **Deception:** None publicly documented
- **Platforms:** Astral Foods investor trading update; coverage by The Record, Security Affairs, TechCentral, WATTPoultry, SC Media, Dark Reading, Reuters / CNBC Africa
- **Reach:** Industry-specific (agriculture / poultry / animal-feed sectors); some general business-press coverage in SA

- **Impact:** Sectoral wake-up call for South African and African agribusiness; placed alongside JBS, Maple Leaf Foods, Dole, Mondelez and HKScan as part of a documented global pattern of ransomware against major food producers (167 ransomware intrusions in agriculture in 2023 per Food and Agriculture-ISAC)
- **Disinfo Investigators:** SC Media, Dark Reading, agricultural-sector ISACs, journalists at The Record and Security Affairs
- **Remediation:** Public, prompt disclosure to investors and regulators; minimal threat-actor messaging in public domain
- **Link to Cyberattack:** Any extortion pressure would have depended on the underlying intrusion; the absence of a publicly visible influence campaign reflects either non-payment without leak follow-through or limited threat-actor sophistication

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AF-20

MTN Group cybersecurity incident and pan-African subscriber-data breach

% fit: 0.8

Date	Publicly disclosed 24-25 April 2025; ransom demand subsequently confirmed by the company
Targets	MTN Group Limited, headquartered in Johannesburg and Africa's largest mobile network operator, with over 290 million subscribers across 18 countries in Africa and the Middle East; affected customers in "certain markets" including South Africa
Continent	Africa
Perpetrator	Not publicly attributed; MTN described the actor only as "an unknown third party"
Motivation	Financial extortion through threatened publication of stolen subscriber data, targeting Africa's largest telecoms group

Cyberattack Details

- **Nature:** Unauthorised access to portions of MTN's IT systems, with subscriber data exfiltration; ransom demand confirmed by the company though MTN has not publicly confirmed whether ransomware encryption was deployed
- **Systems Attacked:** Parts of MTN's customer-data environment in specific national operating units; core network, billing systems and financial-services (Mobile Money / MoMo) infrastructure were not breached
- **Data Held:** Limited contact details — names, surnames, mobile numbers — for an undisclosed number of subscribers in affected markets; no evidence of compromise to mobile-wallet accounts, banking details or authentication credentials
- **Availability Impact:** None to core network or services; MoMo and billing remained fully operational throughout
- **Data Exfiltration/Alteration:** Confirmed exfiltration of personal contact information for subscribers in certain markets; precise volume undisclosed
- **Estimated Impact:** Reputational damage to Africa's flagship telecoms group; identity-theft and phishing risk for affected subscribers; sector-wide regulatory attention across multiple jurisdictions; significant attention to the resilience of African telecoms infrastructure
- **Investigators:** MTN internal cybersecurity teams, South African Police Service (SAPS), Directorate for Priority Crime Investigation ("Hawks"), South African Information Regulator, and counterparts in other affected countries
- **Remediation:** Activation of cybersecurity response processes; notification of affected customers; guidance to subscribers on fraud alerts, MFA, password practices, and avoiding suspicious links; engagement with law enforcement and regulators

Disinformation Details

- **Nature:** Standard ransom-demand pressure typical of data-extortion operators; no public leak-site posting has been confirmed at the time of disclosure
- **Narratives:** Implicit pay-or-publish coercion; the third party "claimed to have accessed data linked to parts of our systems" with the threat that wider release could follow
- **Deception:** None publicly documented beyond standard extortion mechanics
- **Platforms:** MTN press release and investor disclosures; downstream coverage by Reuters / Zawya, BleepingComputer, News24, TechCentral, Security Affairs, Nairametrics
- **Reach:** Pan-African and global financial-press coverage given MTN's size and role
- **Impact:** Heightened attention to the growing pattern of African telecoms breaches (Cell C 2024, Telecom Namibia by Hunters International December 2024, MTN April 2025) and to the value of subscriber data to extortion operators; sector-wide cybersecurity benchmarking pressure

- **Disinfo Investigators:** MTN corporate communications, Check Point Software's threat-intelligence reports on African telecoms, journalists at TechCentral and BleepingComputer
- **Remediation:** Customer-protection guidance; ongoing law-enforcement and regulator engagement; transparency with affected national authorities
- **Link to Cyberattack:** Any extortion pressure depended directly on the cyber intrusion and on credible evidence of held subscriber data

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Asia

This table classifies 14 incidents targeting Asian entities according to the cyber-influence taxonomy:

- **(1) Cyber-enabled influence** — cyber operations serving an influence goal (1a–1h)
- **(2) Influence-enabled cyber** — disinformation/deception enabling a cyber intrusion (2a–2h)
- **(3) Parallel cyber-influence** — simultaneous, independent cyber and influence operations by the same actor
- **1.0** — textbook fit (defining act of the sub-type clearly present).
- **0.7–0.9** — strong fit, partial or mixed on one element.
- **≤ 0.5** — capped here for Type-1 incidents without a real unauthorised intrusion or DDoS, and for Type-2 incidents lacking a tactically-connected subsequent intrusion or DDoS by the same actor.
- **Type-3** scored on whether the cyber and influence components are by the same threat actor and close in time.

Incidents included (% fit ≥ 0.7): 8 of 14

ID	Incident	Country/Scope	Primary Type	Secondary Type(s)	Perpetrator	Victim Category	% Fit
AS-1	BadBazaar surveillance and influence pressure targeting Uyghur populations	China / Uyghur diaspora in Asia and abroad	2c – deepfake/app-impersonation scam (trojanised apps masquerading as trusted community tools)	1c – downstream surveillance and intimidation supporting narrative control	China/PRC-linked (APT15-linked reporting)	Civil society, ethnic-minority activists	0.9
AS-3	Iranian Cyber-Enabled Influence Operations in Support of Hamas	Israel; Bahrain, UAE, US-linked supporters	1e – credit-for-hack (Iranian-affiliated personas amplifying real and inflated breach/wiper claims)	1b – hack-and-leak claims; 1d – hack-and-destroy (data-deletion / disruptive attacks including a US water-authority outage); 3 – parallel attacks during Gaza war	Iran (MOIS-linked group and Hezbollah cyber units)	Government, critical infrastructure, private sector, public opinion	0.9
AS-4	Iranian Cyber Repression and Smear Campaigns Against Ethnic Minorities and Civil Society	Iran (Baluchi, Kurdish, Turkic activists)	1c – hack-and-post (hacked accounts used to spread fake content defaming activists)	2b – phishing and quishing; 1g – DDoS/SIM-disabling for comms suppression of journalists; 1h – impeding activist communications	Iran (Iranian security agencies)	Civil society, journalists, ethnic-minority activists	0.9
AS-5	Lazarus fake cryptocurrency applications targeting users in Asia	Asia (cryptocurrency users)	2c – deepfake/brand-impersonation scam (fake "BloxHolder" app and brand fronts)	2d – website / app impersonation	DPRK (Lazarus Group)	Private sector / individuals (crypto users)	1.0
AS-7	PRC-affiliated defacement and digital-signage hacks during Nancy Pelosi's Taiwan visit	Taiwan	1f – website defacement (and digital-signage hacks at university, 7-Eleven, train stations)	1c – hack-and-post (compromised public displays used as propaganda channels)	China/PRC-linked (PLA Strategic Support Force-affiliated)	Government, public-facing infrastructure, public opinion	1.0
AS-8	PRC cyber and disinformation pressure following Taiwan's 2016 presidential election	Taiwan	3 – parallel cyber-influence (offensive cyber intrusions on government/party sites paralleled by coordinated disinformation barrage)	1f – website targeting	China/PRC-linked actors	Government, ruling party, elections, voters	0.8
AS-9	PRC-linked disinformation over the Osaka rescue narrative and cyber manipulation before Taiwan's 2018 local elections	Taiwan	1c – hack-and-post (DPP party website altered to embed manipulative messaging)	1f – website alteration / defacement-style content insertion	China/PRC-linked actors	Elections, ruling party, public opinion	0.9
AS-14	Arup Hong Kong deepfake-CFO video-conference fraud	Hong Kong (UK-headquartered Arup)	2c – deepfake impersonation scam (live multi-participant deepfake video conference of CFO and colleagues)	2b – initial spear-phishing email pretext from "CFO"; 2g – false-flag impersonation of named executives	Unattributed financially-motivated criminal group with advanced AI tooling	Private sector (engineering / professional services)	1.0

Continental observations

- **Most varied continent in this dataset.** Asia shows the full spectrum: defacements (Taiwan / Pelosi visit), credit-for-hack and disruptive attacks (Iran– Hamas operations), CIB with sham media (Pentagon Iran/Central Asia/MENA campaigns), critical-infrastructure coercive signalling (India power grid), and consumer-scale impersonation scams (Lazarus).
- **Two clear parallel-attack (Type 3) cases:** PRC pressure on Taiwan after the 2016 election and PRC cyber pressure on Indian critical infrastructure — both hybrid-warfare-style.
- **Iran appears as both attacker and victim** (attacker against Israel/US targets; perpetrator-state in repression of its own civil society; target of Pentagon covert campaigns).
- **Notable Western-government attribution** for five incidents — a useful nuance for the keynote on "patriotic hacktivism, disinformation-as-a-service and state-backed cyber warfare coalescing".
- **Arup deepfake-CFO fraud (AS-14)** stands out as the dataset's textbook 2c case: a US\$25.6M loss with no system intrusion at all, achieved entirely through AI-generated impersonation of senior executives during a live video conference.

AS-1

BadBazaar surveillance and influence pressure targeting Uyghur populations

% fit: 0.9

Date	2022
Targets	Uyghur communities in Asia and abroad, including activists and civil-society actors
Continent	Asia
Perpetrator	Chinese-affiliated threat actors, including APT15-linked reporting in the uploaded cyber dataset
Motivation	Surveillance, intimidation, and narrative control over a politically sensitive population

Cyberattack Details

- **Nature:** Mobile surveillance campaign using trojanized applications
- **Systems Attacked:** Android devices via malicious apps masquerading as trusted services
- **Data Held:** Device data, communications, location, and user activity
- **Availability Impact:** No major outage; primary impact was surveillance and compromise
- **Data Exfiltration/Alteration:** Malware enabled collection of device data and user information
- **Estimated Impact:** Persistent surveillance, intimidation, and exposure risk for targeted communities
- **Investigators:** Lookout and dataset compilers
- **Remediation:** App removal, threat advisories, community awareness

Disinformation Details

- **Nature:** The surveillance campaign supported a wider environment of repression and information control affecting Uyghur narratives
- **Narratives:** State-favorable narratives around security and social order; suppression of dissenting accounts
- **Deception:** Trojanized apps masquerading as legitimate tools and community-relevant services
- **Platforms:** Android app ecosystem and digital communications channels
- **Reach:** Targeted but transnational
- **Impact:** Chilling effect on expression and trust in digital tools
- **Disinfo Investigators:** Security researchers and human-rights analysts
- **Remediation:** Exposure and digital-safety guidance
- **Link to Cyberattack:** The cyber intrusion supported intimidation and control over a targeted information community

AS-3 Iranian Cyber-Enabled Influence Operations in Support of Hamas

% fit: 0.9

Date	October 2023-February 2024
Targets	Israel, Israeli supporters including Bahrain, the UAE, and the United States, and wider online audiences exposed to war-related narratives
Continent	Asia
Perpetrator	Iranian government-aligned cyber groups and affiliated actors, with observed collaboration between an MOIS-linked group and Hezbollah cyber units
Motivation	Support Hamas, undermine Israel and its allies, intimidate adversaries, and amplify anti-Israel narratives during the Gaza war

Cyberattack Details

- **Nature:** Targeted hacks, destructive and disruptive cyberattacks, data deletion claims, IoT and industrial-control-related attacks, and operational collaboration across Iranian-linked groups
- **Systems Attacked:** Israeli organizations and supporters, including infrastructure and private-sector targets; one attack took a Pennsylvania water authority offline via Israeli-built PLCs
- **Data Held:** Claimed stolen or deleted data from targeted organizations, operational network access, and system-control pathways relevant to disruption campaigns
- **Availability Impact:** Real operational disruption occurred in some cases, including offline effects at a U.S. water authority and attacks against Israeli-linked systems
- **Data Exfiltration/Alteration:** Iranian-affiliated personas claimed leaks from U.S. companies and took credit for earlier deletion attacks; the broader pattern included both disruptive and leak-style activity
- **Estimated Impact:** Microsoft reported a sharp surge in operations, with cyber-enabled influence campaigns jumping to 11 in October 2023 alone and Iranian groups broadening attacks across several countries
- **Investigators:** Microsoft Threat Intelligence and related security reporting
- **Remediation:** Public warnings, attribution reporting, infrastructure defense, and monitoring of collaboration between Iranian and Hezbollah-linked cyber actors

Disinformation Details

- **Nature:** Cyber-enabled influence operations combining hacks with coordinated social-media amplification and war-related propaganda
- **Narratives:** Support for Hamas, delegitimization of Israel, intimidation of Israel's allies, and framing attacks as retaliation or proof of adversary weakness
- **Deception:** Inauthentic personas, opportunistic social-media amplification, false or inflated claims around leaks and destructive attacks, and possible use of native Arabic speakers to improve persona authenticity
- **Platforms:** Social media, leak claims, and online propaganda channels
- **Reach:** Extended beyond Israel to countries and organizations Iran viewed as supporters of Israel, including Bahrain, the UAE, and the U.S.
- **Impact:** Heightened psychological pressure, widened geographic scope of wartime influence operations, and increased narrative reach tied to cyber incidents
- **Disinfo Investigators:** Microsoft Threat Intelligence and secondary security reporting
- **Remediation:** Threat-intelligence reporting, public exposure of influence patterns, and defense against both cyberattacks and narrative exploitation

- **Link to Cyberattack:** The influence activity was built around targeted hacks and disruptive cyber incidents that were then amplified online to intimidate, shape perceptions, and magnify strategic effect

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AS-4

Iranian Cyber Repression and Smear Campaigns Against Ethnic Minorities and Civil Society

% fit: 0.9

Date	Second half of 2024
Targets	Ethnic minority activists, civil-society actors, journalists, political dissidents, and online communities in Iran, including Baluchi, Kurdish, and Turkic activists
Continent	Asia
Perpetrator	Iranian security agencies and associated cyber-repression actors
Motivation	Suppress dissent, tighten state control over cyberspace, discredit activists, and degrade the online resilience of civil society and minority communities

Cyberattack Details

- **Nature:** Account breaches, phishing, malware, identity fraud, quishing, SIM disabling, device confiscation, and access to online accounts following arrests
- **Systems Attacked:** Activists' accounts, devices, SIMs, messaging groups, and broader digital communications used by civil society and ethnic minority communities
- **Data Held:** Personal information, online-account access, communications, devices seized during arrests, and content-control pathways on Telegram and other platforms
- **Availability Impact:** Disabling of SIM cards and account access disrupted journalists' and activists' ability to communicate and publish
- **Data Exfiltration/Alteration:** Personal data and account access were obtained through breaches, confiscation, and coercive access; fake content was then spread to defame activists
- **Estimated Impact:** Significant restrictions on freedom of expression and systematic pressure on minorities and dissidents; approximately 17.5% of reported cases in the cited dataset involved cyberattacks on Baluchi, Kurdish, and Turkic activists
- **Investigators:** Miaan Group and Filterwatch through the Digital Security Helpdesk dataset
- **Remediation:** Incident documentation, digital-security assistance, exposure of cyber-repression patterns, and support to threatened communities

Disinformation Details

- **Nature:** State-backed smear and content-manipulation campaigns accompanying cyber repression
- **Narratives:** Activists and dissidents as security threats, discrediting of minority voices, and suppression of oppositional narratives through fake or manipulated content
- **Deception:** Fake content planted or spread to defame activists, misuse of Telegram groups, mass reporting to remove accounts or posts, and identity fraud against targets
- **Platforms:** Telegram and other online account ecosystems used by activists and journalists
- **Reach:** Focused on Iranian domestic civic space and minority communities rather than broad mass audiences
- **Impact:** Blended cyber intrusion and psychological pressure to isolate targets, damage reputations, and restrict political expression
- **Disinfo Investigators:** Miaan Group and Filterwatch
- **Remediation:** Digital security support, reporting, documentation, and international awareness of cyber-repression tactics

- **Link to Cyberattack:** The smear campaigns depended on hacked accounts, seized devices, coerced access, and manipulated content pathways created through cyber repression

References

- [1] Miaan Group / Filterwatch, 2024 Iran Cyber Threat Intelligence Report: The Silent War Against Ethnic Minorities and Civil Society — <https://miaan.org/2024-iran-cyber-threat-intelligence-report/>

AS-5 **Lazarus fake cryptocurrency applications targeting users in Asia**

% fit: 1.0

Date	2022
Targets	Cryptocurrency users and digital-finance participants in Asia, including users in Asian markets
Continent	Asia
Perpetrator	North Korean Lazarus Group
Motivation	Financial theft and covert access, supported by deception through fake brands

Cyberattack Details

- **Nature:** Malware distribution through fake cryptocurrency applications
- **Systems Attacked:** User devices and digital-finance environments via trojanized applications
- **Data Held:** Credentials, crypto-wallet access, and device information
- **Availability Impact:** No major outage; the main impact was compromise and theft risk
- **Data Exfiltration/Alteration:** Malware enabled theft of money and cryptocurrency and compromise of user systems
- **Estimated Impact:** Financial loss risk and broader trust erosion in digital-finance tools
- **Investigators:** Volexity and other security researchers
- **Remediation:** Threat warnings, app removal, security monitoring

Disinformation Details

- **Nature:** Deception via a fake brand ("BloxHolder") and trusted-looking app fronts
- **Narratives:** Legitimacy and safety of the fake applications
- **Deception:** Brand impersonation and fraudulent app identity
- **Platforms:** Fake cryptocurrency applications and related online promotion channels
- **Reach:** International, including Asian users
- **Impact:** Users were deceived into self-installing malware
- **Disinfo Investigators:** Security researchers
- **Remediation:** Public exposure and app takedowns
- **Link to Cyberattack:** The cyber compromise depended on deceptive information and branding tactics to lure victims

AS-7

PRC-affiliated defacement and digital-signage hacks during Nancy Pelosi's Taiwan visit

% fit: 1.0

Date	2022
Targets	National Taiwan University, 7-Eleven store displays, train-station displays, Taiwanese public
Continent	Asia
Perpetrator	PLA Strategic Support Force-affiliated actors in public reporting
Motivation	Humiliate Taiwan's government, intimidate the public, and amplify anti-U.S./pro-unification messaging

Cyberattack Details

- **Nature:** Website defacement and digital-signage compromise
- **Systems Attacked:** University website, store display systems, train-station signage systems
- **Data Held:** Public-facing content and display infrastructure
- **Availability Impact:** Public-facing systems were altered and disrupted
- **Data Exfiltration/Alteration:** Display content and website content were altered to carry political messages
- **Estimated Impact:** Public alarm, humiliation of institutions, and visible demonstration of cyber reach
- **Investigators:** Taiwanese authorities, journalists, and researchers
- **Remediation:** Restoration of systems, incident response, public reassurance

Disinformation Details

- **Nature:** Cyber psychological operations using hacked public-facing systems to spread propaganda
- **Narratives:** "There is only one China," anti-Pelosi messaging, anti-U.S. and anti-Taiwan sovereignty narratives
- **Deception:** Use of trusted public infrastructure to present hostile political messages
- **Platforms:** Hacked university website, 7-Eleven displays, train-station displays
- **Reach:** Highly visible national reach
- **Impact:** Fear, humiliation, and message amplification during a sensitive geopolitical event
- **Disinfo Investigators:** Taiwanese authorities, analysts, and media
- **Remediation:** System restoration and public communication
- **Link to Cyberattack:** The influence effect was carried directly through cyber-compromised display and website systems

AS-8

PRC cyber and disinformation pressure following Taiwan's 2016 presidential election

% fit: 0.8

Date	2016 and after
Targets	Taiwanese government, Democratic Progressive Party, Taiwanese voters, Taiwan's online information space
Continent	Asia
Perpetrator	PRC-linked cyber and information actors
Motivation	Pressure Taiwan politically, undermine confidence in the incoming government, and reinforce Beijing's sovereignty claims

Cyberattack Details

- **Nature:** Offensive cyber operations, website targeting, and digital intrusion pressure against Taiwanese institutions
- **Systems Attacked:** Government and party websites, online political infrastructure
- **Data Held:** Government and political communications; exact data categories not enough info
- **Availability Impact:** Not enough info on major outages in the earliest phase of the campaign
- **Data Exfiltration/Alteration:** Website alteration and intrusion pressure were reported; exact exfiltration details not enough info
- **Estimated Impact:** Persistent pressure on Taiwan's democratic institutions and information environment
- **Investigators:** Researchers and public reporting on PRC cyber operations against Taiwan
- **Remediation:** Incident response, public warnings, resilience measures, and counter-disinformation efforts

Disinformation Details

- **Nature:** Coordinated disinformation barrage following Tsai Ing-wen's election win
- **Narratives:** Taiwan is part of China, anti-DPP themes, distrust in Taiwan's leadership
- **Deception:** Spam campaigns, manipulated online discourse, and coordinated messaging
- **Platforms:** Facebook and Taiwan's digital discussion spaces
- **Reach:** National
- **Impact:** Pressure on public confidence and normalization of Beijing's framing
- **Disinfo Investigators:** Researchers, journalists, and policy analysts
- **Remediation:** Digital resilience and public communication efforts
- **Link to Cyberattack:** The disinformation surge accompanied broader PRC cyber pressure against Taiwan's institutions

AS-9

PRC-linked disinformation over the Osaka rescue narrative and cyber manipulation before Taiwan's 2018 local elections

% fit: 0.9

Date	2018
Targets	Taiwanese public opinion, diplomatic credibility, DPP political infrastructure, Taiwanese election discourse
Continent	Asia
Perpetrator	PRC-linked disinformation and cyber actors
Motivation	Undermine trust in Taiwan's government and influence the 2018 election environment

Cyberattack Details

- **Nature:** Website alteration and cyber psychological operations
- **Systems Attacked:** Democratic Progressive Party website and related online political infrastructure
- **Data Held:** Public-facing party content; not enough info on internal data affected
- **Availability Impact:** Not enough info on service outages
- **Data Exfiltration/Alteration:** Website content was altered to display manipulative messaging
- **Estimated Impact:** Reputational damage and political disruption during the election period
- **Investigators:** Taiwanese reporting, researchers, and public analysts
- **Remediation:** Site restoration, public attribution, counter-disinformation response

Disinformation Details

- **Nature:** False narratives about stranded Taiwanese travelers allegedly rescued by the PRC in Osaka
- **Narratives:** PRC competence versus Taiwan government failure, anti-government and anti-DPP framing
- **Deception:** False rescue claims, manipulated political messaging, and cyber-enabled website alteration
- **Platforms:** PTT, social media, altered party website
- **Reach:** National
- **Impact:** Deepened distrust in Taiwan's government and contributed to a severe public backlash
- **Disinfo Investigators:** Journalists, researchers, Taiwanese analysts
- **Remediation:** Fact-checking, public explanation, cyber restoration
- **Link to Cyberattack:** The website alteration directly embedded the political narrative into cyber-compromised infrastructure

AS-14

Arup Hong Kong deepfake-CFO video-conference fraud

% fit: 1.0

Date	January 2024; Hong Kong police notified by Arup in January-February 2024; publicly attributed to Arup by name in May 2024
Targets	Arup, a London-headquartered global engineering and design consultancy with over 18,000 employees across 34 offices (designer of the Sydney Opera House, the Bird's Nest Stadium and many other landmark structures); specifically a finance worker in Arup's Hong Kong office
Continent	Asia (Hong Kong was the locus of the fraud and the financial transfers; Arup is UK-headquartered)
Perpetrator	Unidentified financially-motivated criminal group with access to advanced deepfake video and audio synthesis tools; no arrests as of early 2025
Motivation	Direct theft of corporate funds via AI-enabled impersonation of senior executives, bypassing standard executive-authorisation controls

Cyberattack Details

- **Nature:** AI-enabled social-engineering fraud combining initial spear-phishing email with a multi-participant deepfake video-conference call; no unauthorised intrusion into Arup's systems
- **Systems Attacked:** Arup's financial-authorisation processes and Hong Kong office banking workflows; no internal systems were technically compromised
- **Data Held:** Publicly available source material (LinkedIn videos, YouTube recordings, earnings-call footage) was scraped to build deepfake models of Arup's UK-based CFO and several other colleagues
- **Availability Impact:** None operationally; the impact was entirely financial
- **Data Exfiltration/Alteration:** Not applicable in the technical sense; the "exfiltration" was 15 separate wire transfers totalling HK\$200 million (~US\$25.6 million) to five Hong Kong bank accounts controlled by the perpetrators
- **Estimated Impact:** Direct loss of approximately US\$25.6 million — one of the largest documented deepfake-enabled financial frauds; significant reputational, governance and audit-control consequences; subsequent industry-wide reassessment of executive-authorisation procedures
- **Investigators:** Hong Kong Police Force; Arup internal investigation team led by CIO Rob Greig; later case-study analysis by the World Economic Forum, CNN, Financial Times and Fortune
- **Remediation:** Internal review of financial-controls; multi-channel out-of-band verification mandated for high-value transactions; CIO Rob Greig publicly shared lessons learned; no arrests or fund recovery reported as of early 2025

Disinformation Details

- **Nature:** Direct, targeted AI-driven deepfake impersonation of senior executives — the disinformation operation is the entire intrusion vector
- **Narratives:** "The CFO needs you to process a confidential transaction"; the multi-person video call provided social proof that overwhelmed the employee's initial skepticism
- **Deception:** (1) Initial spear-phishing email impersonating the UK-based CFO requesting a "secret transaction"; (2) deepfake video and audio of the CFO and several colleagues built from publicly available media, used in a live or recorded video conference; (3) authority-based pressure leveraging the perceived seniority of the impersonated executives
- **Platforms:** Email (initial pretext); video-conferencing platform (deepfake call); banking channels (15 transfers to 5 Hong Kong accounts)

- **Reach:** Targeted at a single employee, but the public disclosure created international media reach and serves as a case study in nearly every major analysis of AI-enabled fraud
- **Impact:** Demonstration that deepfakes can defeat traditional video-verification heuristics ("I saw them on camera"); catalysed industry-wide reconsideration of authentication for financial authorisations
- **Disinfo Investigators:** Hong Kong police, Arup internal investigators, CNN Business, Financial Times, Fortune, CFO Dive, World Economic Forum (Rob Greig interview), academic analyses (West Oahu, others)
- **Remediation:** Public awareness campaign by Arup's CIO; new training materials across the industry; corporate adoption of multi-channel verification for large transfers
- **Link to Cyberattack:** The "cyber" component is the deepfake-and-phishing impersonation itself; the influence operation and the cyber operation are wholly fused

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Australia / Oceania

This table classifies 4 incidents targeting Australian entities according to the cyber–influence taxonomy:

- **(1) Cyber-enabled influence** — cyber operations serving an influence goal (1a–1h)
- **(2) Influence-enabled cyber** — disinformation/deception enabling a cyber intrusion (2a–2h)
- **(3) Parallel cyber–influence** — simultaneous, independent cyber and influence operations by the same actor
- **1.0** — textbook fit (defining act of the sub-type clearly present).
- **0.7–0.9** — strong fit, partial or mixed on one element.
- **≤ 0.5** — capped here for Type-1 incidents without a real unauthorised intrusion or DDoS, and for Type-2 incidents lacking a tactically-connected subsequent intrusion or DDoS by the same actor.
- **Type-3** scored on whether the cyber and influence components are by the same threat actor and close in time.

Incidents included (% fit ≥ 0.7): 0 of 4

No incidents meet the % fit ≥ 0.7 threshold for this region.

Continental observations

- Only four incidents are captured, reflecting both the smaller media footprint and Australia's relatively well-defended democratic information environment, but the typology is varied.
- **AU-2 is one of the cleanest examples in the entire dataset of 2e (AI-generated identities)** as a supporting layer to a Type-1c influence operation — useful illustrative material for a 2026 keynote on AI-enabled hybrid threats.
- **Policy framing is itself notable:** AU-1 captures Australia's elevation of misinformation to a *critical-infrastructure* risk, a useful data point for the keynote's question on how governments are responding.
- China-linked activity dominates explicitly attributed Australian cases.

South America

This table classifies 7 incidents targeting South American entities according to the cyber–influence taxonomy:

- **(1) Cyber-enabled influence** — cyber operations serving an influence goal (1a–1h)
- **(2) Influence-enabled cyber** — disinformation/deception enabling a cyber intrusion (2a–2h)
- **(3) Parallel cyber–influence** — simultaneous, independent cyber and influence operations by the same actor
- **1.0** — textbook fit (defining act of the sub-type clearly present).
- **0.7–0.9** — strong fit, partial or mixed on one element.
- **≤ 0.5** — capped here for Type-1 incidents without a real unauthorised intrusion or DDoS, and for Type-2 incidents lacking a tactically-connected subsequent intrusion or DDoS by the same actor.
- **Type-3** scored on whether the cyber and influence components are by the same threat actor and close in time.

Incidents included (% fit ≥ 0.7): 0 of 7

No incidents meet the % fit ≥ 0.7 threshold for this region.

Continental observations

- **South America is the dataset's clearest example of commercial and domestic-political influence operations. Estraterra (SA-3, SA-4) is a textbook case of disinformation-as-a-service*: a commercial PR firm contracted across at least six Latin American election environments — directly relevant to the keynote's question on disinformation-as-a-service.*
- **Almost the entire South American record is Type-1c** — coordinated inauthentic behaviour on Facebook, Instagram, Twitter, TikTok and WhatsApp. No hack-and-leak (1b), defacement (1f), or ransomware-driven cases (2f) are documented in this regional sample.
- **Perpetrator mix is distinctive:** domestic political networks and commercial PR firms (Brazil, Estraterra, Venezuela) dominate, with extra-regional state actors (Russia, China, Pentagon) as a secondary layer riding the same infrastructure.
- **Brazil's 2022 institutional attacks (SA-2)** are notable for explicitly targeting an *electoral institution* (the Superior Electoral Court) — a pattern that prefigures many later attempts elsewhere to delegitimise vote-counting bodies.
- **SA-5 (Pentagon Entorno Diario)** mirrors the AS-10/11/12/13 and EU-27 Pentagon-linked gc_ network operations, confirming this as a globally distributed Western covert influence ecosystem worth flagging in the keynote.

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